MBA Thesis

Beyond Contracts: How AI-Integrated Enterprise Ecosystems Generate Switching Costs and Vendor lock-in

A Qualitative Case Study of AI Infrastructure Platforms

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The author(s) declare(s) that they have completed the thesis work independently. All external sources are cited and listed under the References section. The thesis work has not been submitted in the same or similar form to any other institution(s) as part of another examination or degree.

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Abstract

Enterprise AI adoption is mediated by consolidated cloud platforms integrating compute infrastructure, foundation models, orchestration, and ecosystems. These platforms accelerate implementation but embed vendor dependence transcending contractual lock-in, reaching into technical and operational processes. The mechanisms through which AI platforms generate switching costs remain under-examined.

This qualitative case study examines how enterprise AI platform ecosystems generate vendor dependence and how enterprises respond. The research addresses two questions: (1) What architectural design choices and ecosystem governance mechanisms create switching costs in enterprise AI infrastructure platforms? (2) How do enterprise firms navigate switching cost exposure in enterprise AI infrastructure platforms?

The study employs a Yin Type 2 embedded single-case design with AWS and Azure as supply-side units. Supply-side analysis draws on documentary evidence and independent regulatory and analyst reports to investigate three propositions (P1 - P3). Demand-side analysis from four pilot interviews explores three propositions (P4 - P6); findings are treated as candidate patterns warranting validation.

Documentary analysis identifies six switching cost mechanisms: cross-layer technical dependencies; ecosystem-level lock-in above individual services; ecosystem governance; value-creating features structurally inseparable from dependencies; learning effects compounding through usage; and compliance dependencies amplified by regulatory frameworks. The key finding is structural inseparability of value creation and dependency generation - features delivering highest value produce deepest lock-in.

Three patterns emerge: architectural mitigation develops in response to experienced dependency rather than anticipation; multi-provider approaches function as negotiating instruments; and formal exit governance lags integration depth. Human capital and organisational knowledge accumulation may exceed technical switching costs in practical significance. Compliance frameworks intended to promote portability operationally amplify the structural switching costs they are designed to constrain.

The study advances switching cost theory and platform economics by extending classical analysis to multi-layer AI infrastructure, characterizing lock-in as embedded in value creation rather than contractual restriction, and identifying structural incumbent advantages that current regulatory tools cannot address.

Keywords: AI Platform competition, switching costs, lock-in, AI governance, AI digital platform ecosystem.

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List of abbreviations

AI	Artificial Intelligence
RQ	Research Question
SDK	Software Development Kit. A toolbox that helps developers build applications for a specific platform or service

API	Application Programming Interface. It is a set of rules that lets one program talk to another program
MLOps	The practice of running machine-learning models reliably in real business system
ISV	Independent Software Vendor
Multi-homing	More than one platform at the same time for the same or similar purpose (Like using Azure for some workloads and AWS for others)
PTU	Provisioned Throughput Unit (Azure-specific). A capacity-reservation pricing mechanism that guarantees AI model throughput in exchange for financial commitment
EDP	A Microsoft-level portfolio discount tied to minimum spending commitments across services
IAM	Identity and Access Management. Systems controlling who can access which resources under what conditions.
FinOps	Cloud Financial Operations. Organisational practices for managing, forecasting, and optimising cloud spending
DR	Disaster recovery
Egress (Fees)	Charges applied when data leaves a cloud provider's environment
Throughput	The volume of requests or inferences a system can process over time.
Agent (Ai Agent)	An AI-driven system that performs tasks autonomously by calling tools, APIs, or workflows
AI Sovereignty	The ability for nations or organizations to control the development, deployment and use of AI based on geographical boundaries
Hyperscaler	Large scale cloud service provider with very large data centres spread across the globe

Notational conventions:

In this thesis, the terms 'supply side', 'provider side', and 'platform side' are used interchangeably to refer to AWS and Microsoft Azure. The terms 'demand side', 'enterprise side', and 'buyer side' are used interchangeably to refer to the enterprise organizations adopting these platforms.

1. Introduction

Enterprise AI infrastructure platforms are increasingly becoming core components of organisational IT and data strategy. While these platforms can accelerate model development, deployment, and scaling, they can also generate switching costs through tightly coupled architectures, proprietary tooling, data and workflow dependencies, and ecosystem-based complementarities. When these switching costs accumulate to the point where customer mobility is structurally constrained, they produce vendor lock-in - a state of platform dependency in which migration to an alternative provider becomes economically, technically, or operationally infeasible within the relevant decision horizon. Vendor lock-in reduces strategic flexibility, increases long-term costs, and limits the enterprise's ability to respond to changing technological or regulatory conditions.

Industry analysis suggests that the basis of competition in cloud markets is gradually shifting away from basic infrastructure services, which are becoming more standardised across providers, and towards AI-enhanced offerings that play a growing role in shaping enterprise platform choices; the same analysis projects that enterprises which do not adapt their underlying AI compute environment may face cost differences of more than 50 percent by 2030 (Gartner, 2025). Early platform adoption decisions may therefore form the basis of long-term commitments (Farrell and Klemperer, 2007).

Prior research has examined vendor lock-in in cloud migration (Opara-Martins et al., 2016) and broader platform competition dynamics (Jacobides et al., 2018; Rong et al., 2026). However, it has rarely integrated provider-side mechanisms with enterprise-side strategic responses in the specific context of enterprise AI infrastructure platforms. These provider-side mechanisms include architectural design choices such as platform openness strategies (Boudreau, 2010) and ecosystem governance (Costabile, 2024; Helfat and Raubitschek, 2018).Existing research examines switching costs primarily in cloud computing or general platform contexts, without accounting for the multi-layer dependencies introduced by AI-integrated infrastructure platforms (Hagiu and Wright, 2025), and tends to treat architectural design, ecosystem governance, and enterprise strategy in isolation. This creates substantive uncertainty for decision-makers and a corresponding theoretical gap regarding how provider mechanisms and enterprise responses jointly produce lock-in outcomes in this setting.

Switching costs, as used in this study, refer to the economic, technical, organisational, and learning-related costs incurred by an enterprise when changing from one AI infrastructure platform to another. These include compatibility constraints, data migration challenges, accumulated organisational learning, contractual dependencies, and ecosystem-specific complementarities, but exclude short-term price fluctuations. Ecosystem governance refers to the rules, incentives, access conditions, pricing structures, and coordination mechanisms through which platform providers shape participation, value creation, and dependency within their AI infrastructure ecosystems.

This study makes three contributions to Industrial Economics and Management. First, it extends Switching Cost Theory (Klemperer, 1995; Farrell and Klemperer, 2007; Besanko et al., 2017) to AI-integrated enterprise platforms, a setting where multi-layer technical, organisational, and ecosystem dependencies operate simultaneously across a vertical stack that classical single-layer formulations do not directly anticipate. Second, it provides mechanism-level evidence linking provider-side architectural and governance choices to enterprise-side switching cost outcomes, extending the established analysis of ecosystem complementarities and platform interdependencies (Jacobides et al., 2018; Evans and Schmalensee, 2016) to the AI infrastructure context. Third, it offers a practitioner-facing framework for evaluating platform dependency risk and designing mitigation strategies across platform selection, governance negotiation, and multi-cloud diversification decisions.

This study focuses primarily on the platform layer of enterprise AI infrastructure, where value-creating AI services and ecosystem governance mechanisms intersect to generate switching costs. On the

supply side, the study analyses two embedded platform units - Amazon Web Services and Microsoft Azure - selected as the dominant configuration of enterprise AI platform provision globally (Synergy Research Group, 2026). On the demand side, the study examines enterprise organisations operating within these ecosystems through pilot interviews with enterprise decision-makers. Findings are generalised analytically - from case to theory - rather than statistically from sample to population..

2. Purpose and research question(s)

The purpose of this study is to examine how architectural design choices and ecosystem governance mechanisms in enterprise AI infrastructure platforms generate switching costs, and how enterprise firms navigate the switching cost exposure during AI platform adoption.

Research Questions

RQ1: What architectural design choices and ecosystem governance mechanisms create switching costs in enterprise AI infrastructure platforms?

RQ2: How do enterprise firms navigate switching cost exposure in enterprise AI infrastructure platforms?

Each research question addresses one side of the provider-enterprise interaction that jointly produces switching costs. RQ1 examines the supply-side mechanisms - multi-layer cost activation, the embedding of switching costs in value-creating features, and ecosystem governance dynamics. RQ2 examines the enterprise-side dynamics - the trade-off between platform value and lock-in exposure, within-user learning effects that compound switching costs through normal usage, and the persistence of structural switching costs under regulatory intervention.

Two research questions are necessary because switching costs in enterprise AI infrastructure platforms arise from a joint dynamic: providers structure mechanisms on the supply side, and enterprises respond strategically on the demand side. Examining only one side would mischaracterize lock-in as either purely structural or purely behavioural. A single research question would therefore risk conflating platform-imposed constraints with user-driven strategic responses, obscuring the interaction through which switching costs are produced and sustained. It would either over-attribute causality to provider design or to enterprise agency, failing to capture how the two are mutually constitutive. RQ1 addresses the supply side; RQ2 addresses the demand side.

3. Previous research

The chapter reviews existing research related to switching costs, vendor lock-in, digital platform ecosystem governance, and multiproduct platform strategies. The literature review was conducted using a structured search strategy to ensure transparency and reproducibility. *The primary academic databases used were **Scopus** and **Summon** and supplemented with additional searches in **Google Scholar**, **ScienceDirect**, and **ResearchGate.** The search combined predefined keywords related to enterprise AI platforms, switching costs, vendor lock-in and ecosystem governance, using Boolean operators to refine results. The review covered literature from 2000-2026 and exceptional reference to dynamic competition (1988,1995) to switching cost with emphasis on more recent literature (post-2018) to reflect the rapid evolution of enterprise AI infrastructure and cloud ecosystems and applying inclusion criteria focused on peer-reviewed, English-language studies on cloud platforms, AI infrastructure, and platform dependency.*

The objective is to establish a theoretical foundation for examining how architectural design choices and ecosystem governance mechanisms in enterprise AI infrastructure platforms generate switching costs, and how enterprise firms navigate the trade-off between platform value capture and switching cost exposure. While much of the literature originates from cloud computing and digital platform studies, these perspectives provide valuable insights into the emerging dynamics of enterprise AI platforms.

Switching costs constitute a central theoretical concept in platform economics and competition. Klemperer (1995) and Farrell and Klemperer (2007) provide the canonical theoretical framework, demonstrating that switching costs operate as a structural source of market power that allows incumbents to sustain advantages even in the presence of equally capable competitors. Burnham, Frels and Mahajan (2003) operationalise these costs through a tripartite typology that distinguishes financial switching costs (loss of financially quantifiable resources), procedural switching costs (time and effort of moving), and relational switching costs (psychological discomfort and the breaking of bonds). Although their empirical work was conducted in B2C consumer settings, the typology has been widely adopted across switching cost research and provides a classificatory scheme that this study extends to enterprise AI platform contexts. Pick and Eisend's (2014) meta-analytic assessment of switching cost antecedents confirms that the typology operates robustly across studies, with alternative attractiveness, prior product/provider experience, and relationship quality producing consistent effects on switching behaviour. He et al. (2023) extends this analysis to platform settings specifically, examining the interaction between switching costs and network externalities and demonstrating that high switching costs can weaken competitive pressures and reinforce platform dominance. Their two-stage model shows that when switching costs are relatively high and network externalities relatively low, incumbent platforms adopt closed strategies that harvest rents from their installed base, which is a dynamic observable in enterprise AI platform pricing structures.

Vendor lock-in has long been recognised as a significant challenge in enterprise technology adoption, particularly within cloud computing environments. Opara-Martins, Sahandi and Tian (2016) conceptualise vendor lock-in as a condition in which the cost of switching providers exceeds the perceived benefits, driven by proprietary technologies, data migration challenges, and contractual dependencies. Their study emphasises that lock-in is not solely a technical issue but also a business risk that affects strategic flexibility, cost control, and long-term organisational autonomy. Although cloud platforms offer scalability and innovation advantages, they simultaneously embed enterprises within tightly coupled service ecosystems.

Besanko et al. (2017, pp. 364-365) identify switching costs as a strategic variable operating on both sides of the supplier-buyer relationship. On the buyer side, substantial costs arise when firms develop brand-specific know-how: expertise built around a particular system's interfaces, workflows, and

operational logic. This constitutes a sunk investment losing value upon switching. On the supplier side, sellers can deliberately design their products to increase switching costs through loyalty programmes, tied warranties, and product bundling where complementary goods are designed to function together, making replacement of any single component costly without disrupting the whole. This distinction between organically accumulated switching costs and architecturally engineered switching costs maps directly onto the central question of which design choices and governance mechanisms generate lock-in in enterprise AI platforms.

3.1. Ecosystem Governance and Multiproduct Platform Strategies

The importance of ecosystem governance in shaping platform dynamics is well established in the literature. Costabile (2024) provides a systematic literature review of digital platform ecosystem governance, identifying key governance building blocks such as access rules, control mechanisms, pricing structures, and coordination processes, and arguing that governance choices influence participation, value distribution, and power relationships among platform stakeholders. In enterprise AI contexts, governance mechanisms shape access to models, data services, deployment pipelines, and compliance tools, and therefore represent an important source of switching costs that operate alongside technical architecture. Chen et al. (2022) further demonstrate that governance and design in digital platforms are deeply intertwined, with incentive and control mechanisms routinely shaping how platform participants create and appropriate value, which is a dynamic directly applicable to how AI platform providers structure enterprise dependency.

Helfat and Raubitschek (2018) extend this governance perspective by linking ecosystem design explicitly to value capture. They argue that platform leaders deploy three types of dynamic capabilities - innovation, environmental scanning, and integrative capabilities for ecosystem orchestration, not only to create value but also to capture a disproportionate share of the value generated within the ecosystem. The integrative capability dimension is particularly relevant for enterprise AI platforms, where pricing structure and governance structure are intertwined in ways that shape both complementor incentives and enterprise switching costs.

Engert et al. (2025) examine self-organisation and governance in digital platform ecosystems through an information ecology lens, demonstrating that effective governance combines centralised orchestration with mechanisms that permit decentralised adaptation by ecosystem participants. Their findings illustrate that platform governance is not a static design choice but a dynamic process in which platform leaders continuously calibrate the balance between control and complementor autonomy. This dynamic is directly observable in how AWS and Azure structure access tiers, marketplace curation and partner programmes.

Boudreau (2010) provides empirical evidence on how distinct platform-opening approaches produce qualitatively different innovation and dependency dynamics. Studying 21 handheld computing systems, Boudreau finds that granting access to a platform and opening markets for complementary components produces up to a fivefold acceleration in complementor activity, while devolving control over the platform itself produces a positive but order-of-magnitude smaller effect. This distinction between access-granting and control-devolving approaches is directly relevant to enterprise AI platforms, where AWS's open-but-integrated architecture and Azure's exclusive-access architecture represent two distinct platform-openness configurations with correspondingly different switching cost profiles.

Kretschmer, Leiponen, Schilling and Vasudeva (2022) frame platform ecosystems as meta-organisations that span traditional firm boundaries, arguing that strategic management research must develop new frameworks to capture how platforms coordinate independent actors without formal hierarchical authority. This meta-organisational framing complements Jacobides et al.'s (2018) ecosystem theory and is directly applicable to enterprise AI platforms, where AWS and Azure

coordinate dense networks of independent software vendors, foundation model providers, and enterprise customers through governance instruments rather than vertical integration. Jacobides, Cennamo and Gawer (2024) extend this framework further. They demonstrate that the same complementarities that generate ecosystem value can move from structural solutions to endogenous failures, locking both platforms and complementors into dependencies that neither party can unilaterally exit. This finding is directly relevant to the bilateral specificity dynamic this study examines in enterprise AI platforms.

Platform providers increasingly adopt multiproduct strategies that bundle complementary services into integrated ecosystems (Jacobides et al., 2018; Evans and Schmalensee, 2016). Rong et al. (2026) describe this in the contemporary AI context as an all-in-one platform strategy that crafts multiproduct ecosystems to enhance customer value while strengthening retention through compounding interdependencies (developed further in Section 3.2). Hagiu and Wright (2025) provide the structural framework for understanding what is distinctive about AI-integrated cloud ecosystems: the vertical AI stack. Here, across hardware, cloud, data, foundation models, and applications, layered dependencies are created where dominance at one layer can be leveraged across others, extending the classical ecosystem analysis of Jacobides et al. (2018) to a multi-layer setting. An enterprise migrating AI workloads cannot do so by addressing any single layer in isolation, for switching costs operate at multiple layers simultaneously. Hagiu and Wright (2025) further note that each large player in the AI stack has incentives to commoditise the layers where it is weak. Meta, for example, supports open-source LLMs to avoid a foundation model monopoly, and major technology companies develop proprietary chips to reduce Nvidia dependency. This creates strategic opportunities for enterprises to play providers against each other wherever commoditised alternatives exist.

3.2. Platform Value Creation and the Lock-in Trade-off

A recurring theme across the platform economics and industrial organisation literature is that the mechanisms through which platforms create value for enterprise users are structurally identical to the mechanisms through which they generate switching costs. This relationship is not incidental. It is the defining strategic characteristic of integrated platform ecosystems, and it creates a fundamental trade-off that enterprise firms must navigate when making AI platform adoption decisions.

Besanko et al. (2017, pp. 364-365) establish the theoretical foundation for this trade-off. The bilateral specificity introduced in the chapter opening combines buyer-side brand-specific know-how with supplier-side value-creating structures designed to make benefits inseparable from retention mechanisms. It implies that the enterprise capturing the most value from deep integration is simultaneously the enterprise facing the highest switching cost upon exit. The trade-off is not a flaw of platform design but its defining strategic feature.

Evans and Schmalensee (2016, pp. 188-199) extend this argument to multi-sided platform competition, identifying four forces through which platforms disrupt traditional markets: an improved way of doing things, a radically different pricing model, indirect network effects that incumbents cannot duplicate, and global reach that incumbents cannot replicate alone. Each of these value-creating forces simultaneously deepens user dependency. The improved capability that attracts enterprise adoption is the same proprietary service that creates compatibility costs. The radically different pricing model that reduces upfront investment is the same consumption structure that penalises exit through committed-use discount forfeiture. The indirect network effects that enrich the platform ecosystem are the same externalities whose value cannot be transferred to a competing provider.

Cennamo and Santaló (2019), drawing on collective action theory, demonstrate that the value-creation/dependency dynamic also operates within the broader ecosystem. They show that generativity in platform ecosystems produces a tension between system-reputation benefits - which accrue collectively to all complementors - and free-riding incentives that erode the quality of complementary

innovation as platforms mature. This dynamic reinforces the structural argument that platform value creation and lock-in are inseparable, because the collective ecosystem value that attracts enterprises is the same value that becomes non-transferable upon exit.

Rong et al. (2026) demonstrate this dynamic at the ecosystem level through the all-in-one platform strategy. Providers that craft multiproduct ecosystems to enhance enterprise value simultaneously construct interdependencies that make partial exit structurally irrational. An enterprise cannot extract maximum value from an integrated AI platform while simultaneously maintaining full optionality to exit, because the value derives precisely from the depth of integration. Jacobides et al. (2018) reinforce this point at the ecosystem architecture level, arguing that the complementarities and interdependencies through which ecosystems generate value for participants are the same structural features that create asymmetric bargaining power between platform orchestrators and enterprise adopters over time.

This dynamic is particularly acute in enterprise AI infrastructure contexts. The capabilities delivering the highest operational value, such as managed AI services with superior model access, integrated MLOps environments, and platform-native compliance infrastructure, are the same capabilities that generate the deepest technical, organisational, and financial switching costs. An enterprise cannot simply avoid lock-in by refusing to use these capabilities, because those capabilities are the reason it chose the platform in the first place. The strategic dilemma is therefore not between using a platform and avoiding lock-in, but between the degree of value capture and the degree of dependency exposure and hence a trade-off that must be actively managed rather than avoided.

3.3. Platform Learning Effects and Within-User Dependency

A critical distinction for understanding how lock-in deepens over time (between across-user and within-user learning effects) is introduced by Hagiu and Wright (2025) and extends the established analysis of learning-based switching costs (Swann, 2009, pp. 84-85; Burnham et al., 2003) to the AI platform context. Across-user learning creates data network effects, whereby more users improve the product for all by generating data that enables AI improvement across the entire user base. Within-user learning, by contrast, creates compounding switching costs: as a given enterprise continues to use a platform, the platform becomes increasingly customised to that enterprise's patterns, preferences, and workflows, making the enterprise progressively less inclined to switch to a competitor that lacks this accumulated knowledge. This distinction explains why enterprise lock-in deepens autonomously through normal usage. As a consequence, as the platform learns the enterprise's operational patterns, the cost of moving to a platform that lacks this learned knowledge rises continuously without any deliberate lock-in action by either party.

Burkhardt and Rieder (2024) extend this dynamic to the foundation model layer, arguing that foundation models constitute a new class of platform model in which prompting practices, fine-tuning workflows, and prompt-engineering know-how generate platform-specific dependencies that have no precise analogue in traditional cloud computing. The accumulated prompt libraries, fine-tuning datasets, and model-calibration practices that an enterprise builds within a specific provider's foundation model environment constitute a platform-specific organisational asset. This asset is non-portable, value-generating, and progressively deepening with use, and it operates as a within-user learning mechanism in the sense identified by Hagiu and Wright (2025).

Swann (2009, pp. 84-85) provides complementary grounding for this dynamic. Even in the absence of brand loyalty, a firm may be reluctant to switch to an objectively superior alternative if the cost of doing so outweighs the expected benefit. This reluctance stems from familiarity with the existing product, the sunk costs in implementation and configuration, and accumulated expertise and relational capital that would need to be rebuilt from scratch with a new provider. Swann (2009) further notes that switching costs alone do not produce complete reluctance to move. They raise the threshold for

switching but do not eliminate it entirely. It is when switching costs combine with network effects that technological lock-in becomes particularly severe, with each force reinforcing the other.

3.4. Portability, Regulation, and the Limits of Interoperability

Emerging regulatory frameworks, including the European Data Act and the EU AI Act mandate portability and interoperability with the stated objective of reducing vendor lock-in and enabling competitive market entry. However, the literature suggests that such interventions address the contractual and data-format surface of switching costs while leaving structural and architectural costs largely intact. Opara-Martins et al. (2016) establish that cloud lock-in is a business risk extending far beyond the technical layer, affecting strategic flexibility and organisational autonomy in ways that data portability mandates are not designed to resolve. Hagiu and Wright (2025) similarly argue that traditional barriers to entry from switching costs and ecosystem lock-in persist even as regulatory frameworks target more visible mechanisms such as data formats and API interoperability standards.

Schrepel and Pentland (2024) extend this regulatory analysis to the foundation model layer, arguing that competition between AI foundation models is shaped by training cost barriers, GPU concentration, and proprietary toolchain dependencies that operate above and beyond the contractual and data-format dimensions current regulatory frameworks target. Their analysis reinforces the broader finding that structural ecosystem characteristics like vertical integration across the AI stack, network effects, and proprietary toolchain dependencies reinforce market concentration in ways that interoperability requirements alone are unlikely to dissolve.

The OECD (2025) identifies three categories of switching barrier in AI infrastructure that are particularly resistant to regulatory remedy. The first is contractual and pricing structures (multi-year volume commitments, exclusivity clauses, and egress fees) that limit short-term mobility. The second is the absence of agreed standards across the AI supply chain, which makes switching suppliers operationally costly on a scale. The third is proprietary software ecosystems tied to specific hardware, such as Nvidia's CUDA, which create network effects that make a complete ecosystem switch the only viable migration path. The OECD (2025) notes that these barriers are not necessarily the result of illegal conduct but add structural friction to competition. This is directly consistent with this study's argument that the most significant switching costs are embedded in value-creating features rather than deliberately restrictive mechanisms. Empirical corroboration is provided by the UK Competition and Markets Authority's (2025) cloud services market investigation, which conducted primary data collection from AWS and Microsoft Azure and concluded that competition in the UK cloud market is not working well, with both providers holding *„significant unilateral market power "*(CMA, 2025, para. 36). The CMA found that less than 1% of customers switch cloud provider each year (CMA, 2025, para. 24), and identified egress fees, technical barriers, and Microsoft's licensing practices as features giving rise to adverse effects on competition (CMA, 2025, para. 36). The investigation's empirical conclusion that committed spend agreements *„in their current form and application... do not harm competition "*(CMA, 2025, para. 28) while egress fees and technical barriers do, supports the present study's argument that the most consequential lock-in operates through structural rather than purely contractual mechanisms.

3.5. Synthesis and Research Motivation

Prior research establishes switching costs as a key source of platform power but provides limited insight into how AI-specific architectural choices intensify these costs across multiple stack layers simultaneously. Ecosystem governance is well-theorised conceptually yet underexplored empirically in enterprise AI infrastructure settings. The interaction between platform learning effects and enterprise lock-in remains insufficiently examined at the organisational level. These gaps motivate RQ1, which examines provider-side architectural and governance mechanisms, and RQ2, which focuses on how enterprise firms navigate the trade-off between platform value capture and switching

cost exposure. The practical urgency of these questions is reflected in industry projections that platform lock-in will rise from 5 percent to 35 percent of organisations by 2027 (Gartner, 2026), suggesting that the mechanisms studied here are intensifying rapidly rather than stabilising.

Table 1 in Appendix C synthesises the preceding review by mapping each identified mechanism to its position in the AI stack, the type of switching cost it generates, its enterprise implication, and the proposition it motivates.

The streams of research reviewed above generate the following six propositions for empirical investigation through a qualitative case study. Propositions P1 through P3 address the supply side - the platform-level mechanisms through which switching costs are created. Propositions P4 through P6 address the demand side - how enterprise firms try to manage the resulting dependencies.

Propositions for RQ1 - Supplier / Platform-Side: Creation of Switching Costs

P1: AI integration into enterprise cloud platforms activates multiple switching cost types simultaneously (compatibility, learning, contractual, data, and ecosystem) such that aggregate switching costs exceed the sum of individual components due to cross-layer dependencies across the vertical AI stack.

P2: The architectural design choices that create the most significant switching costs are embedded in the platform's value-creating features such as managed AI services, integrated development environments, marketplace ecosystems rather than in deliberately restrictive mechanisms such as egress fees or proprietary data formats.

P3: Ecosystem governance mechanisms, including pricing structures that discourage multi-homing, marketplace curation that deepens complementor dependencies, and abstraction that hides infrastructure complexity while increasing technical dependency generate switching costs that operate at the ecosystem level rather than the individual service level.

Propositions for RQ2 - Enterprise-Side: Managing switching costs

P4: Enterprise strategies for managing switching costs involve a trade-off between platform value capture and lock-in exposure, where the features delivering the greatest operational value typically generate the highest switching costs.

P5: Within-user learning effects cause enterprise switching costs to compound autonomously through normal platform usage, such that lock-in deepens over time without deliberate action by either the platform or the enterprise.

P6: Despite regulatory intervention mandating portability and interoperability, structural switching costs in AI-integrated cloud ecosystems persist because they are embedded in technical architecture and organizational processes, not merely in contractual terms.

4. Theory

4.1. Switching Cost Theory

This study adopts Switching Cost Theory as its single primary analytical lens. Switching costs, defined here as the economic, technical, organisational, and learning-related costs incurred when an enterprise changes from one platform provider to another are the central theoretical construct through which the empirical phenomenon is examined. Three supplementary theoretical perspectives - network effects and platform economics, ecosystem governance, and transaction cost economics are drawn on selectively to extend or contextualise switching cost analysis where the empirical setting requires it. These supplementary perspectives are not applied in parallel with the primary lens; they are tools used in service of the switching cost analysis.

This study draws on two strands of switching cost theory. The canonical formulations of Klemperer (1995) and Farrell and Klemperer (2007) establish switching costs as a structural source of market power. The bilateral specificity framework of Besanko et al. (2017, pp. 364–365) then positions them as a co-created dependency embedded in the supplier–buyer relationship. The Burnham, Frels and Mahajan (2003) tripartite typology (financial, procedural, relational) provides the operational scheme this study extends to enterprise AI contexts, with Pick and Eisend's (2014) meta-analytic confirmation of its robustness. Swann's (2009, pp. 84-85) baseline argument - that switching costs raise the threshold for switching but do not eliminate it and combine with network effects to produce technological lock-in - anchors the enterprise-level interpretation of these mechanisms. Farrell and Shapiro (1988) provide the theoretical grounding for how committed-capacity and discount-forfeiture pricing structures operate as switching cost instruments, directly applicable to the commercial commitment mechanisms in enterprise AI contexts. Yang et al. (2025, working paper) apply a related framework to the AWS-Azure competitive context specifically, modelling leader-follower dynamics under high switching cost conditions. As an unreviewed working paper this is treated as a contemporary illustration rather than a foundational source. The substantive review of these sources is presented in Chapter 3. Together these formulations establish switching costs as a structural source of market power that this study applies and extends to the multi-layer context of enterprise AI platforms.

Limitations of Switching Cost Theory

Switching cost theory has three limitations relevant to its application in this study. First, the canonical formulations (Klemperer, 1995; Farrell & Klemperer, 2007) and much of the empirical literature (Burnham et al., 2003; Pick & Eisend, 2014) were developed in B2C consumer markets, where switching is typically a discrete individual decision. Enterprise AI platform switching is a collective organisational decision shaped by multiple stakeholders, governance processes, and political dynamics that the theory does not directly address. Second, the theory assumes broadly informed buyers who can perceive and evaluate switching costs ex ante; in enterprise AI contexts, pricing abstraction, technical opacity, and the within-user learning effects identified by Hagiu and Wright (2025) cause switching costs to be partially obscured at the moment of decision and to compound autonomously thereafter. Third, the theory was developed in product-market settings where switching cost mechanisms were largely separable from value-creating mechanisms. In modern integrated platform contexts and particularly in AI-integrated infrastructure, value creation and switching cost generation may be structurally inseparable, a configuration the original theoretical formulations did not anticipate. Whether and how the theory needs to be extended to accommodate this inseparability is one of the analytical questions this study addresses. These limitations do not undermine the theory's analytical utility but condition how it is applied: switching cost theory is treated here as a generative analytical framework that this study applies and seeks to extend to the enterprise AI infrastructure context, rather than as a settled doctrine to be tested unmodified.

4.2. Supplementary Theoretical Perspectives

The following perspectives are drawn on selectively to extend the primary switching cost lens where the empirical setting requires it.

Network Effects and Platform Economics: Evans and Schmalensee (2016, pp. 188-199), He et al. (2023) and Hagiu and Wright (2025) provide the framework for network effects in multi-sided platform competition. The value-creating forces of these platforms, including superior capability, distinctive pricing, indirect network effects, and global reach, simultaneously deepen user dependency. High switching costs and network externalities then interact to reinforce platform dominance. Hagiu and Wright (2025) extend this to the vertical AI stack, characterising cross-layer lock-in dynamics and introducing the distinction between across-user learning which creates data network effects and within-user learning, which causes switching costs to compound autonomously through normal usage. This perspective extends switching cost theory by explaining how network effects and learning dynamics amplify switching cost mechanisms in AI platform settings (substantive review in Sections 3.2 and 3.3). It supports P1, P3, and P5 and is applied in Sections 6.4.1 and 6.4.3.

Ecosystem Governance: Costabile (2024), Chen et al. (2022), Helfat and Raubitschek (2018), Jacobides et al. (2018), and Rong et al. (2026) provide the conceptual vocabulary for the rules, access conditions, pricing structures, and complementarities through which platform providers shape switching costs at the ecosystem level. This perspective extends switching cost theory by addressing the ecosystem-level layer of analysis that the primary lens alone does not fully capture (substantive review in Section 3.1). It supports P3 and is applied in the analysis of marketplace integration, partner program stratification, and complementor-side switching costs in Sections 6.2.3, 6.3.3, and 6.4.1.

Transaction Cost Economics: Transaction cost economics (TCE) provides a complementary lens for understanding why switching costs accumulate and persist at the organisational level. Enterprise relationship-specific investments in platform integration (customised workflows, proprietary toolchain configurations, fine-tuned models) constitute asset specificity in the transaction cost sense, and learning effects and data feedback loops in AI-intensive settings continuously deepen this asset specificity, transforming discrete switching costs into persistent transaction costs embedded in the enterprise's operational architecture. This perspective is particularly relevant to P5 and is applied in the analysis of within-user learning and progressive asset specificity in Sections 6.2.5, 6.3.5, and 6.4.

4.3. Conceptual Framework

The literature reviewed in Chapter 3 surfaces five mechanisms that link platform architecture and ecosystem governance to enterprise switching costs and motivate the propositions developed below.

First, compatibility and learning costs arising from proprietary APIs, toolchains, and platform-specific know-how constitute the primary technical switching cost mechanism, motivating P1 and P2. **Second**, multi-homing disincentives, access tiers, marketplace curation, and architectural bundling create ecosystem-level switching costs that operate above the individual service level and cannot be addressed by replacing any single platform component, motivating P3. **Third**, the features delivering the greatest enterprise value generate the highest switching costs because value creation and dependency generation are structurally inseparable in integrated AI platform ecosystems, motivating P4. **Fourth**, within-user learning effects cause switching costs to compound autonomously through normal platform usage, deepening lock-in over time without deliberate action by either provider or enterprise, motivating P5. **Fifth**, regulatory interoperability mitigates contractual and data-format switching costs but does not address switching costs embedded in technical architecture, organisational processes, and compliance infrastructure, motivating P6. These five mechanisms structure the conceptual framework and provide the analytical bridge between the literature reviewed in Chapter 3 and the propositions tested empirically in Chapter 6.

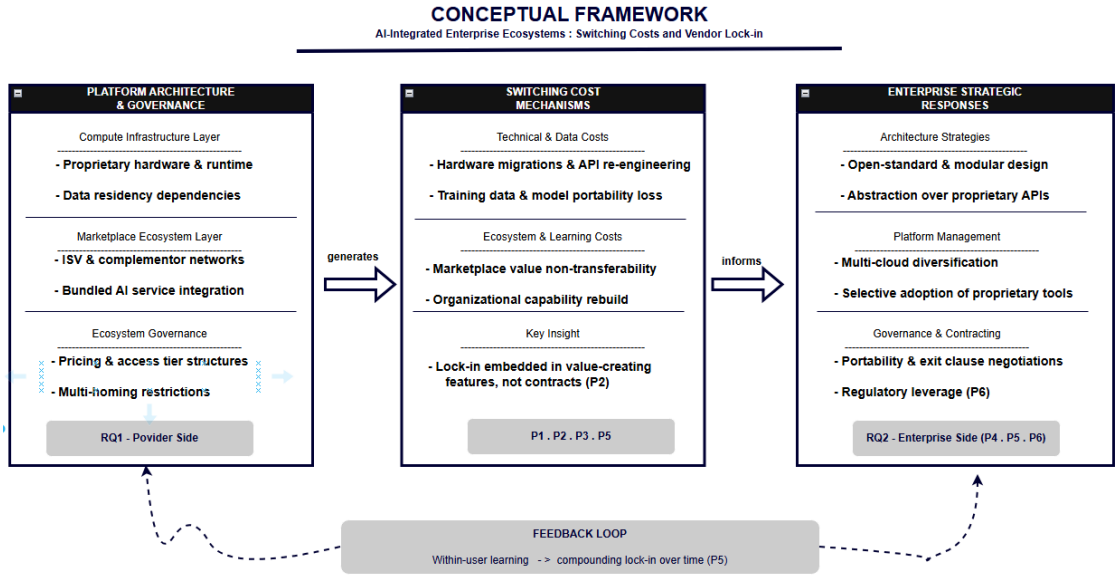


Figure 1 - Switching costs and vendor lock-in conceptual framework

The conceptual framework illustrates how switching costs and vendor lock-in may emerge within AI-integrated enterprise platforms and how enterprise navigate the trade-off between platform value capture and switching cost exposure. The framework is structured around three core components - platform architecture and governance, switching cost mechanisms, and enterprise strategic responses connected through a reinforcing feedback loop, as shown in Figure 1.

On the provider side, platform architecture and ecosystem governance represent the proposed primary sources of switching costs. Architectural choices such as proprietary runtimes, managed AI services, and tightly integrated development environments may create technical dependencies, while ecosystem governance mechanisms (pricing structures, access tiers, and marketplace arrangements) shape how enterprises interact with complementary services and partners. Together, these design and governance choices form the structural conditions under which switching costs are proposed to arise, addressed by RQ1. These platform-level conditions may generate switching cost mechanisms that operate across technical, organizational, and ecosystem dimensions. Technical and data-related costs may arise from migration complexity, API re-engineering, and limited portability of trained models and data pipelines. At the ecosystem level, switching costs may emerge through the non-transferability of marketplace value and the need to rebuild organizational capabilities and workflows when changing platforms. A central proposition of this framework is that these switching costs are embedded in value-creating platform features rather than driven primarily by contractual restrictions (P2).

On the enterprise side, firms are proposed to respond to these switching costs through strategic management choices addressed by RQ2. These may include architectural strategies such as modular design and abstraction over proprietary interfaces, platform management strategies such as multi-cloud diversification and selective use of proprietary services, and governance strategies involving contract negotiation, portability clauses, and regulatory leverage. Enterprise strategies are proposed to reflect a trade-off between capturing platform value and limiting long-term dependency (P4). The framework further incorporates a feedback loop driven by within-user learning effects. As enterprises continue to use a platform, organizational knowledge, customization, and data accumulation may deepen platform-specific integration, causing switching costs to compound over time (P5). This dynamic is proposed to explain how vendor lock-in can intensify even in the absence of deliberate restrictive actions by the platform provider, which is an empirical question this study addresses through the six propositions developed in Chapter 3.

5. Research Methodology

This study adopts a qualitative research strategy grounded in an interpretivist epistemological position. The phenomenon under investigation - how architectural design choices and ecosystem governance mechanisms generate switching costs, and how enterprises manage these costs strategically is constituted through the decisions and contextual judgements of platform architects and enterprise technology leaders. Understanding it requires preserving institutional and relational complexity, not abstracting it away. Three conditions, as identified by Yin (2014, p. 4), determine the appropriateness of case study as the research strategy for this study. First, both research questions are process-oriented: RQ1 asks what mechanisms exist and how they operate; RQ2 asks how enterprise firms navigate switching cost exposure in enterprise AI infrastructure platforms. These are characterisation and process questions that require depth and context, not frequency measurement. Second, the researcher exerts no control over the architectural choices made by platform providers or the procurement decisions of enterprise firms - ruling out experimental designs. Third, the phenomenon is contemporary and deeply embedded in its institutional context: AI-specific switching costs are inseparable from the contractual structures, API architectures, and ecosystem dynamics that constitute enterprise AI platforms. Yin (2014, p. 16) identifies this condition through identifying where „the boundaries between phenomenon and context are not clearly evident "as the defining case for case study research. This is further supported by Eisenhardt (1989, p. 532), who argues that case study is particularly suited to research aimed at theory building from empirical evidence, especially where constructs are grounded in varied contextual data and where the goal is extending or refining existing theoretical frameworks. This study does precisely that. It applies switching cost theory to the novel context of enterprise AI infrastructure platforms, drawing on the canonical formulations of Klemperer (1995) and Farrell and Klemperer (2007) and the later developments of Besanko et al. (2017), Hagiu and Wright (2025), and Swann (2009). It then extends the theory by characterising mechanisms that operate across the vertical AI stack, which the canonical formulations do not directly address.

Research Design

This study adopts a Type 2 embedded single-case design (Yin, 2014, p. 50). The case is defined as the phenomenon of switching cost generation and strategic management in enterprise AI infrastructure platform ecosystems - a bounded system studied in its real-world context. AWS and Microsoft Azure are not the cases; they are embedded supply-side sub-units of analysis, selected because they represent the dominant configuration of enterprise AI platform competition globally (Synergy Research Group, 2026) and exhibit meaningful variation in architectural philosophy and governance structure. This variation enables characterization of both platform-generic and platform-specific mechanisms without constituting separate cases, as the analytic focus remains the ecosystem-level phenomenon rather than the firms per se. The selection of this ecosystem as the single case is justified on representative and revelatory grounds (Yin, 2014, p. 52), given its global dominance and the previously undertheorized nature of AI-specific switching dynamics. The design (Visualized more in table 2) contains two types of embedded unit, each mapped to a research question: Sub-unit findings are synthesised upward to illuminate the overarching case rather than compared laterally. This preserves the Type 2 design logic and prevents drift toward a multiple-case structure (Yin, 2014, pp. 54-55).

The methodology chapter is organized as follows: data collection method (5.1), sampling (5.2), ethical considerations (5.3), method of analysis (5.4), credibility and validity (5.5), approach which shows the step-by-step research procedure (5.6).

5.1. Data Collection method

This study employed a multi-source evidence strategy on the supply side, complemented by primary interview evidence on the demand side. Yin (2014, pp. 120 - 123) identifies multi-source documentary

evidence as a defining principle of rigorous case study research because it enables triangulation: the convergence of independent evidence sources toward common findings, strengthening construct validity. On the supply side, multiple document types pricing schedules, contractual terms, technical documentation, and institutional reports were used to triangulate findings. On the demand side, the pilot interview sample was treated as exploratory primary evidence, with triangulation operating across the supply-side and demand-side evidence types at the case level rather than within each unit type.

Type A - Supply-side documentary analysis: Supply-side evidence was collected from publicly available primary sources documenting the architectural and commercial structure of AWS and Microsoft Azure AI platforms. The documentary corpus comprised three categories of source.

Pricing schedules: Amazon Bedrock, Amazon SageMaker AI, Azure AI Foundry Models, Azure AI Services commitment tiers, Azure OpenAI Service, and Microsoft Foundry.

Contractual and legal terms: AWS Customer Agreement, AWS Service Terms, AWS Responsible AI Policy, and Microsoft Product Terms for Online Services.

Technical and governance documentation: Amazon Bedrock and SageMaker AI documentation, AWS IAM User Guide, AWS Marketplace Seller Guide, Azure Resource Manager and Azure Machine Learning documentation, Azure Marketplace Publisher Guide, and Microsoft Cloud Partner Programme documentation.

Institutional and industry analysis from OECD (2025), the UK Competition and Markets Authority (2025), Gartner (2025; 2026), Forrester Research (2025), and Synergy Research Group (2026) was used to corroborate provider-specific findings and to anchor the empirical setting against independent regulatory and market analysis. All documentary sources were accessed and archived during the data collection period to ensure the analysis reflected a stable evidentiary base; the access date for each source is recorded in the reference list.

Type B - Demand-side pilot interviews: Four semi-structured pilot interviews were conducted with enterprise informants meeting the criteria described in Section 5.2. The semi-structured format provided a consistent set of topics derived from the theoretical propositions in Chapter 4, while allowing informants to introduce themes not anticipated in the protocol - a balance between comparability and discovery appropriate given the exploratory character of RQ2 (Yin, 2014, p. 110). Interviews were conducted remotely, recorded with informed consent, transcribed verbatim, and analysed in NVivo alongside the documentary corpus. The interview guide and template letters for the interview are included in Appendix A and B respectively.

5.2. Sampling

Selection of both supply-side units and demand-side informants was governed by theoretical rationale rather than statistical representativeness. As Yin (2014, p. 21) distinguishes, case study research seeks analytical generalisation - from case to theory and not statistical generalisation from a representative sample to a population.

Supply side: AWS and Azure were selected on four grounds: First, they represent the dominant configuration of enterprise AI platform provision, with AWS and Microsoft Azure together accounting for approximately 49 percent of global enterprise cloud infrastructure spending in Q1 2026 with AWS at 28 percent and Azure at 21 percent of a USD 128.6 billion quarterly market, making the case representative of the broader market (Synergy Research Group, 2026). Second, industry forecasts indicate that more than 60 percent of enterprises will run intensive AI activity across multiple clouds by 2030, with AWS and Azure consistently named as the leading providers in that segment (Gartner,

2025); the AWS-Azure dyad therefore reflects the empirically relevant multi-platform context. Third, the two platforms vary enough in architecture and governance to support characterization of both common and provider-specific mechanisms. Fourth, both publish extensive documentation, enabling multi-source evidence collection. Studying both providers as embedded units within a single case therefore captures the multi-platform reality enterprises navigate, while preserving the analytical depth associated with embedded single-case designs (Yin, 2014).

Demand side: Enterprise informants were sought against the following criteria: large organisations (500+ employees) with production-scale AI integration on AWS, Azure, or both; informants holding senior roles with direct responsibility for cloud strategy, enterprise architecture, or AI platform governance; and representation across multiple industry sectors to support assessment of whether findings are platform-generic or sector-specific. The original sampling plan targeted four to five pilot interviews across at least three sectors. The realised sample comprised four informants - two in telecommunications, one in banking and one in federal services / management consulting, reflecting the practical access constraints typical of senior technology leader recruitment. The pilot framing was adopted because this sample size, while sufficient to develop initial demand-side evidence in support of analytical generalisation to the theoretical propositions, did not reach saturation in the technical sense (Yin, 2014, p. 21). Informants were recruited through professional networks and referral from contacts, an appropriate access strategy for senior technology leaders who are not publicly enumerable. The implications of the realised sample size and sector coverage for the scope of demand-side findings are addressed in Section 7.4.

Sampling limitations: Sampling-related limitations including potential geographic and sectoral skew of the demand-side sample, possible self-selection effects, and the supply-side restriction to two providers are addressed in detail in Section 7.4 alongside the broader limitations of the study.

5.3. Ethical Considerations

This research adhered to ethical guidelines suitable for studies involving human participants, particularly in semi-structured pilot interviews with enterprise technology decision-makers, alongside the analysis of publicly available documentary sources from cloud service providers.

Informed consent was obtained from all interview participants prior to data collection. Participants were made aware of the nature of the study, its academic purpose, the voluntary character of their involvement, and their right to withdraw at any point without consequence. To uphold anonymity and confidentiality, interviewees were referred to by descriptive role-and-sector codes (e.g., VP-Telco, Architect-FedSvc) and organisations were identified by sector and approximate size only. This precaution was particularly important given the commercially sensitive nature of cloud platform decisions, as respondents discussed contractual terms and internal architecture choices that could carry competitive or professional implications if attributed. Interview recordings, transcripts, and analytical memos were stored in password-protected cloud storage accessible only to the researchers and, where necessary for supervision, the thesis supervisor. The supply-side documentary analysis drew exclusively on publicly available primary sources (pricing schedules, contractual terms, and technical documentation) published by AWS and Microsoft as well as institutional and industry analysis documents. These sources were accessed during the data collection period, archived for analytical use, and cited transparently throughout the analysis. No proprietary or confidential provider information was used.

Finally, no funding, conflicts of interest, or commercial affiliations biased the study. The research was conducted purely as part of an academic MBA programme under the ethical supervision of Blekinge Institute of Technology in accordance with its ethical research guidelines.

5.4. Method of analysis

Operationalisation links the theoretical constructs developed in Chapter 4 to observable empirical indicators. The primary constructs are switching costs, ecosystem governance mechanisms, and enterprise strategic management responses, anchored in the theoretical framework of this study rather than in external measurement instruments. Switching costs were operationalised using the taxonomy developed from Besanko et al. (2017, pp. 364-365), He et al. (2023), and Hagiu and Wright (2025): compatibility costs, learning costs, contractual and financial costs, and ecosystem costs, with the within-user learning dynamic identified by Hagiu and Wright (2025) treated as a cross-cutting mechanism observable across all cost types. Ecosystem governance mechanisms were operationalised following Costabile (2024) and Rong et al. (2026): access rules and tiering, pricing structures that discourage multi-homing, marketplace curation policies, and abstraction layers that hide infrastructure complexity while deepening technical dependency. Enterprise strategic responses to the value/lock-in trade-off (RQ2) were examined through three observable dimensions, treated as preliminary patterns rather than comprehensive coverage: architectural strategies (abstraction layer investment, modular design, portability-by-design); platform management strategies (multi-cloud diversification, selective adoption of proprietary services); and governance strategies (contract negotiation, portability clauses, regulatory leverage). These dimensions were drawn directly from the six propositions developed in Chapter 4 and provided the structure for the demand-side interview.

5.4.1. Method of analysis - Type A (Supply-side documentary analysis)

Documentary analysis applied the proposition-driven strategy described in Section 5.1. Each documentary source was read in full and coded against the analytical codes specified in Appendix C Table 4. Where a passage instantiated a switching cost mechanism for example, a pricing schedule describing a commitment-based discount with non-transferable benefits, it was tagged with the relevant code and linked to the proposition it evidenced. Coding proceeded independently for each embedded unit (Azure, then AWS), with pattern matching performed within each unit before cross-unit synthesis. Mechanisms not anticipated by the propositions were retained as candidate emergent findings, following Yin's (2014, pp. 147-150) explanation building logic. Triangulation across pricing schedules, contractual terms, and technical documentation established that observed mechanisms were consistently structured rather than artefacts of any single document.

5.4.2. Method of analysis - Type B (Demand-side pilot interviews)

Interviews were recorded with informed consent, transcribed verbatim, and coded in NVivo using the same analytical framework as the documentary analysis, with the following A-G-N attribution rules applied to classify informant accounts.

Attribution rules: Architecture, Governance, and Network-effects coding. When informants reported lock-in experiences, the analytical attribution of their account to Architecture (A), Governance (G), or Network-effects (N) followed explicit decision criteria to ensure coding consistency and transparency across both embedded unit types. An informant account was attributed to **Architecture (A)** when two or more technical codes were present simultaneously, such as proprietary API dependency, data-format lock, SDK specificity, or runtime incompatibility and re-implementation effort was the primary reported constraint, with the barrier persisting even if contract terms were renegotiated. Example prompt*: „Which internal systems now depend on the provider's APIs, SDKs, or proprietary data formats, and what would it take to re-engineer them? "*. An account was attributed to **Governance (G)** when policy, contract, or pricing structures bound the enterprise despite technical portability being feasible, that is, where technical alternatives existed but financial exit penalties, minimum spend obligations, or compliance requirements made migration commercially irrational. Example prompt: *„Has a contractual provision - around minimum commitments, termination fees, or data export restrictions - shaped your architecture decisions? "*. An account was attributed to **Network-effects**

(N) when the reported value of the platform depended on cross-user or within-user learning effects accumulated model customisation, workflow encoding, or non-transferable ecosystem complementor value and lock-in deepened autonomously through normal usage rather than through contractual or architectural constraints. Example prompt: *„At what point did you feel your integration became difficult to reverse - and was that driven by a specific decision or by gradual accumulation? "*. Where an informant account contained elements of more than one category, which was expected given P1's proposition that switching costs activate simultaneously across dimensions, all applicable codes were assigned, and the dominant mechanism was identified through the pattern matching against the propositions.

Coding proceeded proposition by proposition, with each transcript reviewed for evidence of P1-P6 and informant accounts tagged against the analytical codes in Table 4. Where an account contained elements of more than one A-G-N category, all applicable codes were assigned, and the dominant mechanism was identified through pattern matching against the propositions. Findings from the pilot interviews were synthesised across informants to identify common patterns, and these patterns were cross-referenced against the supply-side findings during the integration step described below. Given the pilot character of the interview sample, demand-side findings were treated as exploratory propositions warranting further empirical validation, not as saturated empirical claims.

5.4.3. Integration of supply-side and demand-side findings

Following the within-unit analyses, supply-side and demand-side findings were synthesised at the case level through explanation building (Yin, 2014, pp. 147-150). Supply-side findings (the platform-side mechanisms creating switching costs) and demand-side findings (the enterprise-side strategic responses) were aligned proposition by proposition, with attention to convergence (where both sources converge on the same finding), complementarity (where each source illuminates a different facet of the same proposition), and divergence (where the two sources point in different directions, requiring consideration of rival explanations to determine which best accounts for the evidence). Auditability was maintained through a case study protocol documenting all analytical decisions and a case study database in which the supply-side documentary corpus was archived in dated, retrievable form and the demand-side interview transcripts and analytical memos were managed in NVivo, both of which Yin (2014, pp. 84-87) identifies as essential reliability measures for case study research.

5.5. Credibility and validity

Four quality criteria were applied following Yin (2014, pp. 45-49). Construct validity was addressed through multi-source triangulation across documentary types on the supply side, through cross-unit triangulation between supply-side and demand-side evidence at the case level, and through the operationalisation framework in Section 5.4, which grounds all measures in the theoretical framework. Internal validity was addressed through pattern matching against the study's six propositions and through explanation building in which a rival explanation was specified for each proposition and adjudicated against the assembled evidence. This procedure yielded partial support for P5 and a qualified verdict for P6 and retained P1 and P2 only after testing them against specified alternatives, confirming that the design was capable of producing disconfirming as well as confirming outcomes rather than confirming the propositions by construction. External validity was achieved through analytical rather than statistical generalisation: findings were generalised to the theoretical propositions rather than to a population, supported by the representative and revelatory rationales for case selection. Reliability was ensured through a documented case study protocol; a case study database comprising an archived supply-side documentary corpus and demand-side coding with full audit-trail traceability; and verbatim transcription of interview recordings.

Reflexive memos were maintained throughout data collection and analysis, documenting interpretive decisions, potential biases introduced by prior familiarity with the platform domain, and the rationale

for analytical choices made at each stage. These memos form part of the case study database and contribute to the audit trail.

5.6. Approach

This section describes the step-by-step execution of the research procedure across data collection, analysis, and integration phases.

Phase 1 - Theoretical framework development. The research began with a literature review covering switching cost theory as the primary lens, supplemented by perspectives from platform economics, ecosystem governance, and transaction cost economics, leading to the development of six propositions (P1-P6) structuring both the supply-side and demand-side analyses. The propositions were anchored in Klemperer (1995), Farrell and Klemperer (2007), Besanko et al. (2017), Swann (2009), Hagiu and Wright (2025), and supplementary perspectives.

Phase 2 - Supply-side data collection. Documentary sources were identified through systematic search of AWS and Microsoft Azure publicly available pricing schedules, contractual terms, technical documentation, and policy documents. Each source was archived with its access date. Independent institutional and analyst sources (OECD, CMA, Gartner, Forrester, Synergy Research Group) were collected to provide corroboration and empirical setting.

Phase 3 - Demand-side data collection. Four semi-structured pilot interviews were conducted with senior technology leaders identified through professional networks and referral. Each interview was scheduled remotely, conducted in approximately one hour, recorded with informed consent, and transcribed verbatim. Interviewees received the interview guide and consent letter (Appendices A and B) in advance.

Phase 4 - Coding and within-unit analysis. Documentary sources were coded against the analytical framework in Table 4, with coding proceeding independently for each embedded unit (Azure, then AWS). Interview transcripts were coded in NVivo using the same analytical framework, with the A/G/N attribution rules applied to classify informant accounts. Pattern matching was performed within each unit before cross-unit synthesis.

Phase 5 - Cross-unit synthesis and integration. The within-unit analyses of Azure and AWS were compared to identify platform-generic and platform-specific mechanisms. Supply-side findings were then aligned with demand-side findings proposition by proposition - with the supply-side analysis carrying the principal weight of evidence for P1-P3 and the demand-side analysis carrying the principal weight for P4-P6, with attention to convergence, complementarity, and divergence patterns at the integration step.

Phase 6 - Reflexive review and writing. Throughout the analysis, reflexive memos documented interpretive decisions and potential biases. Findings were drafted with continuous reference to the case study protocol and database, ensuring auditability of all analytical claims.

6. Analysis

6.1. Introduction and analytical structure

The analysis is presented in three parts. Part 1 (Sections 6.2 - 6.5) looks at the supply side embedded unit (Type A) and part 2 (Sections 6.6 - 6.9) at the Demand side embedded unit (Type B). Propositions are examined within each sub-unit independently before addressing the case level findings in part 3 (sections 6.10 - 6.11).

PART 1: SUPPLY-SIDE WITHIN-UNIT ANALYSIS (DOCUMENT ANALYSIS)

This section presents the within-unit analysis of the two Type A supply-side embedded units (Microsoft Azure and Amazon Web Services), examining how each platform's pricing architecture, technical design, and ecosystem governance mechanisms generate switching costs in enterprise AI infrastructure contexts. Both platforms are analysed as embedded sub-units within the single overarching case, consistent with the Type 2 embedded single-case design described in Section 5 (Yin, 2014, p. 50). The analytical procedure follows Yin's (2014, pp. 136-137) proposition-driven strategy: Section 6.2 presents the Azure analysis, Section 6.3 the AWS analysis, Section 6.4 the cross-unit synthesis, and Section 6.5 the case-level findings addressing RQ1.

6.2. Within-unit analysis: Microsoft Azure

Microsoft Azure accounted for approximately 21 percent of global enterprise cloud infrastructure spending in Q1 2026 (Synergy Research Group, 2026). Azure's AI pricing is built around two core models - consumption-based and provisioned throughput, augmented by commitment tiers and enterprise purchasing agreements, each introducing distinct forms of financial, operational, and architectural dependency (Microsoft, 2026a; 2026b; 2026e; 2026h; 2026i). Architectural and ecosystem mechanisms are corroborated through Microsoft's publicly available technical and programme documentation (Microsoft, 2026c; 2026d; 2026f; 2026g).

6.2.1. P1 - Multi-Layer Switching Cost Activation

P1 proposes that AI integration activates multiple switching cost types simultaneously such that aggregate switching costs exceed the sum of individual components. This occurs due to cross-layer dependencies across the vertical AI stack, generating a form of super-additive lock-in through complementarities between technical, data, and commercial layers. Azure AI Foundry exemplifies this dynamic. It provides unified access to foundation models from OpenAI, Meta, Mistral, and Cohere alongside Microsoft's Phi family, yet all model inference is executed within Azure-managed infrastructure. Usage is billed through Azure subscriptions, governed by Microsoft Entra ID identity controls, and generates telemetry data stored in Azure Monitor and Azure Log Analytics. As a result, an enterprise using Foundry simultaneously incurs dependencies across the model access, identity, observability, and data layers, forming a coordination nexus in which individual dependencies cannot be disentangled without affecting the system. Azure's commercial structure compounds this at the financial level. Enterprise consumption commitments are negotiated at portfolio level, applying discounts across compute, storage, networking, and AI services simultaneously. Consequently, AI workload migration triggers consequences across the entire Azure relationship. The cross-layer dependency is further reinforced by Azure Resource Manager (ARM) templates, which encode infrastructure-as-code in a Microsoft-proprietary format not directly portable without transformation to other cloud providers (Microsoft, 2026f), and the Azure ML Studio Pipelines API, which encodes ML workflows in Azure-specific definitions (Microsoft, 2026c). An enterprise operating Azure OpenAI Service carries simultaneous dependencies across four layers. At the commercial layer, these are provisioned throughput and enterprise commitments, with consumption tiers typically used for

testing. At the model layer, they are prompt engineering and system behaviour calibrated to specific model families such as GPT-4o. At the data layer, they are training datasets and fine-tuned weights embedded in Azure storage. At the platform layer, they are identity management through Microsoft Entra and monitoring via Azure-native observability services. The cross-unit theoretical analysis of multi-layer switching cost activation is developed in Section 6.4.1.

6.2.2. P2 - Switching Costs Embedded in Value-Creating Features

P2 proposes that the most significant switching costs in enterprise-scale AI deployments are embedded in value-creating features rather than deliberately restrictive mechanisms. Access to GPT-4o and related OpenAI models under enterprise-grade managed service terms is provided through Azure OpenAI Service (Microsoft, 2026e); it is this managed access and not contractual restriction that creates the dependency. Value is delivered through identity, billing, monitoring, compliance, and quota governance simultaneously, such that extracting the capability strips away the performance guarantees that make it valuable. Evans and Schmalensee (2016, pp. 188-199) identify this as the defining competitive characteristic of multi-sided platforms: the features attracting users are the same features making departure costly. Azure Machine Learning Studio evidence P2 at the development tooling layer. Its value in integrated IDEs, automated pipeline orchestration, experiment tracking, one-click deployment etc is also the source of its switching cost. Enterprises accumulate institutional workflows, pipeline configurations, and deployment scripts specific to Azure ML's APIs and SDK conventions. The managed environment that eliminates infrastructure complexity is architecturally identical to the mechanism that makes accumulated work non-portable. Data egress pricing, by contrast, represents a deliberately restrictive mechanism - explicit, transparent, contestable, and increasingly subject to regulatory scrutiny under frameworks such as the European Data Act. The switching costs embedded in managed AI services, enterprise-grade model access, and provisioned throughput commitments operate through value creation and are therefore structurally more durable than egress pricing alone - implicit, endogenous, and difficult to regulate without simultaneously degrading the value they generate. This is consistent with the „beyond contracts “framing of this study: the most consequential lock-in sources are those an enterprise would choose to deepen even if given complete freedom to exit. See Section 6.4.2. for the platform-specific manifestation.

6.2.3. P3 - Ecosystem Governance Mechanisms

P3 proposes that ecosystem governance mechanisms (pricing structures discouraging multi-homing, marketplace curation deepening complementor dependencies, and abstraction hiding infrastructure complexity) generate switching costs at the ecosystem level rather than the individual service level. **Provisioned Throughput Units** (PTUs) illustrate the commitment-based component. PTUs are platform-specific, capacity-reserved pricing instruments that are non-transferable across cloud providers: reserved Azure OpenAI throughput cannot be redeployed on AWS Bedrock or Google Vertex AI. Enterprises committing to PTUs face full on-demand pricing on competing platforms, raising the effective cost of parallel multi-homing at production scale. **Azure's consumption constructs** (tokens, PTUs, commitment credits) do not map cleanly onto pricing units of competing platforms, requiring non-trivial normalisation effort for cross-platform cost comparison. **Azure Marketplace** embeds complementary services within Azure's billing, identity, and procurement infrastructure (Microsoft, 2026d). Enterprises that procure third-party AI and ML tools like data labelling services, vector databases, monitoring platforms, compliance solutions, and domain-specific models through this channel consolidate vendor relationships, identity management, and financial governance within a single platform boundary natively integrated with Microsoft Entra identity. Azure operates a stratified Microsoft **Cloud Partner Programme** structure spanning Solutions Partner, Specialisation, and Microsoft AI Cloud Partner designations (Microsoft, 2026g), tying partner status to revenue thresholds, certification commitments, and technical specialisations. ISVs are incentivised toward Azure-native integration through preferential placement, co-sell programmes, and revenue-

share structures that scale with Azure-billed transaction volume. **Ecosystem governance** through cross-product licensing extends beyond Azure-internal mechanisms. The UK Competition and Markets Authority's (2025) cloud services market investigation found that Microsoft's licensing practices for some of its software products „are adversely impacting the competitiveness of AWS and Google in the supply of cloud services "(CMA, 2025, para. 35). The CMA further found that the input price Microsoft charges to AWS and Google for some products „can be higher than Microsoft's customer-facing price for some cloud customers "(CMA, 2025, para. 31), and that „customers with existing licences cannot bring these to AWS and Google in most instances "(CMA, 2025, para. 33). Enterprises evaluating migration must consider not only the direct switching costs of Azure-native services but also the increased costs of running Microsoft software workloads on another platform.

6.2.4. P4 - Value Capture and Lock-in Exposure

P4 proposes that enterprise strategies for managing switching costs involve a trade-off between platform value capture and lock-in exposure. The supply-side analysis establishes the structural conditions making this unavoidable: platform design embeds value creation and switching cost generation in the same features. How enterprises navigate this trade-off is examined in Sections 6.6 to 6.9. Provisioned throughput pricing exemplifies this structurally. PTUs deliver guaranteed model throughput, predictable cost management, and performance stability which offer operationally significant advantages for production-scale AI workloads. Yet the same structure creates a sunk financial commitment and a platform-specific reservation that cannot be transferred upon exit. The enterprise deriving the greatest operational value by running the highest workloads is the enterprise facing the highest financial switching cost. The enterprise discount programme extends this logic at the portfolio level: deeper discounts require higher minimum spend commitments, and the enterprise with the deepest EDP discounts is simultaneously the one facing the highest exit cost. Azure AI Foundry's provisioned capacity commitments compound this at the AI service layer. Across all three commitment structures, the enterprise capturing the most operational value bears the highest switching cost which points to a pattern structurally designed into the platform's commercial architecture.

6.2.5. P5 - Within-User Learning and Compounding

P5 proposes that within-user learning effects cause enterprise switching costs to compound autonomously through normal platform usage, deepening lock-in over time without deliberate action by either party. Consumption-based pricing tracks and amplifies this accumulation: usage scales with proficiency, so the deeper an enterprise's platform competence, the greater its consumption and operational calibration. Every API call against Azure OpenAI Service and every workflow built against Azure's managed AI services incrementally deepens calibration to Azure's idiosyncratic operational patterns. A further compounding mechanism operates at the organisational process level, below the technical layer. As development workflows, agent behaviours, and task architectures become encoded around Azure-specific tooling, accumulated institutional practice becomes platform-specific in ways that cannot be migrated without substantial retraining. Besanko et al. (2017, pp. 364-365) describe this as brand-specific know-how that becomes a sunk investment losing value upon switching. The use-it-or-lose-it structure of certain consumption commitments creates a ratchet effect as teams normalise higher consumption levels, deepening operational dependency each billing period without any discrete decision point. The cross-unit theoretical analysis of this within-user learning mechanism is developed in Section 6.4.3. Demand-side experience is examined in Sections 6.6 to 6.9.

6.2.6. P6 - Persistence Despite Regulatory Intervention

P6 proposes that despite regulatory intervention mandating portability and interoperability, structural switching costs persist because they are embedded in technical architecture and organisational processes rather than contractual terms. The Microsoft Product Terms for Online Services (Microsoft,

2026i) make explicit what is contractually binding addressing data training restrictions, model weight protections, use restrictions. But the most durable switching costs are not in these contractual terms. Regulatory frameworks such as the European Data Act target legally verifiable mechanisms like data portability, API interoperability, egress fee caps but address only the visible surface of a dependency structure. Regulation is not useless but structurally incomplete. Opara-Martins et al. (2016) establish that cloud lock-in is a business risk extending far beyond the technical layer, implying that interventions targeting technical portability alone are structurally insufficient. Regulatory compliance requirements act as an amplifier of platform dependency. Enterprises in regulated sectors face mandatory data residency, security, and audit requirements that Azure's compliance infrastructure is engineered to satisfy (Microsoft, 2026i documents Azure Government configurations, sovereign cloud environments, and compliance certifications). The regulatory environment does not constrain lock-in - it deepens it by making Azure's compliance infrastructure operationally necessary. Compliance infrastructures are economically indivisible and benefit from scale, so enterprises rationally converge on platforms minimising audit and certification risk; over time compliance becomes infrastructural rather than merely legal. P6's persistence claim therefore applies with more in regulated sectors.

6.3. Within-unit analysis: Amazon Web Services

AWS accounted for approximately 28 percent of global enterprise cloud infrastructure spending in Q1 2026 (Synergy Research Group, 2026). AWS's AI pricing architecture is primarily structured around two core models - on-demand inference pricing and provisioned throughput pricing augmented by additional financial and technical layers, including SageMaker Savings Plans and costs associated with model customisation and fine-tuning (Amazon Web Services, 2026b; 2026d; 2026e; 2026i). Architectural and ecosystem mechanisms are corroborated through technical and programme documentation (Amazon Web Services, 2026a; 2026c; 2026f; 2026g; 2026h). A structural feature warranting attention is the use of abstracted and composite pricing units. Bedrock's token-based billing applies heterogeneous rates across foundation models with each priced independently, while SageMaker pricing spans multiple components across instance classes, regions, and usage durations (Amazon Web Services, 2026a; 2026b; 2026d). This pricing opacity constitutes a structural information asymmetry of the kind Hagiu and Wright (2025) identify as a mechanism through which AI platforms sustain competitive advantage.

6.3.1. P1 - Multi-Layer Switching Cost Activation

Amazon Bedrock provides direct evidence of P1. Bedrock provides unified API access to foundation models, but all model inference runs through AWS-managed infrastructure, is billed through AWS accounts, is subject to AWS IAM identity controls, and generates usage data stored in CloudWatch and CloudTrail. An enterprise using Bedrock simultaneously carries dependencies at the model access, IAM, observability, and data storage layers. Migration requires not merely finding equivalent model access but decoupling from an integrated AWS infrastructure stack spanning multiple services. The SageMaker ecosystem compounds this as it integrates model training, fine-tuning, deployment, monitoring, and evaluation within a single managed environment connecting to Amazon S3, Amazon Redshift, AWS Glue, and Amazon EMR - all services whose pricing and commitment structures interact with SageMaker's. The December 2024 launch of SageMaker Unified Studio deepened this integration by combining ML development with AWS analytics infrastructure, making the AI development environment architecturally inseparable from broader data infrastructure.

6.3.2. P2 - Switching Costs Embedded in Value-Creating Features

Bedrock's value proposition is unified access to multiple frontier models like Claude, Llama, Mistral, Nova - through a single API, unified billing, and consistent security controls. Yet the same unified API that delivers this simplification creates the switching cost: all model inference flows through

Bedrock's API structures, authentication mechanisms, and response formats. An enterprise that has built production workflows against Bedrock's unified interface cannot simply redirect to direct API calls without re-engineering the entire integration layer. SageMaker's managed ML environment evidences P2 at the development tooling layer. Its value in eliminating ML infrastructure management, integrated IDEs, automated training pipelines, one-click deployment is also the source of its switching cost. Enterprises accumulate institutional workflows, notebook configurations, pipeline definitions, and deployment scripts specific to SageMaker's APIs and SDK patterns. These services that solve operational problems are the same mechanisms making accumulated work non-portable.

6.3.3. P3 - Ecosystem Governance Mechanisms

AWS's provisioned throughput pricing structure evidences the multi-homing disincentive. TPU reservations on Bedrock are platform-specific and non-transferable, with minimum enterprise commitments of approximately $15,000 per month. An enterprise maintaining TPU reservations faces the full on-demand price on competing platforms, making meaningful multi-cloud AI inference economically prohibitive at scale. AWS's consumption constructs (tokens, TPUs, Savings Plan commitment hours) do not map cleanly onto competing platforms' pricing units, requiring non-trivial normalisation effort for cross-platform cost comparison. AWS Marketplace embeds complementary services within AWS's billing, identity, and procurement infrastructure (Amazon Web Services, 2026g). Enterprises that procure third-party AI and ML products (data labelling tools, vector databases, ML monitoring platforms, compliance solutions, and specialist domain models) through this channel consolidate vendor relationships, identity management, and financial governance within a single platform boundary natively integrated with AWS IAM identity. AWS operates a stratified AWS Partner Network competency programme, tying partner status to revenue thresholds, technical certifications, and competency designations (Amazon Web Services, 2026g). ISVs are incentivised toward AWS-native integration through preferential placement, co-sell programmes, and revenue-share structures that scale with AWS-billed transaction volume. AWS's ecosystem governance operates through provisioned throughput commitments, marketplace integration, and partner programme stratification rather than through cross-product business-software licensing.

6.3.4. P4 - Value Capture and Lock-in Exposure

SageMaker Savings Plans exemplify the trade-off at the MLOps infrastructure layer. Savings Plans deliver discounts of up to 64% on training and inference costs compared to on-demand rates, applicable across instance families and regions. For enterprises running large-scale AI training workloads, these represent material budget savings. Yet the structure requires a one or more years minimum spend commitment across SageMaker workloads. Enterprises with the greatest savings - running the largest, most consistent workloads face the highest financial penalty for reduction or migration before the term expires. Model customisation pricing compounds this trade-off at the AI capability layer. Enterprises fine-tuning foundation models through SageMaker AI accumulate AI capabilities specifically calibrated to their operational context. The fine-tuned model is more valuable than the base model as it encodes domain-specific knowledge. This accumulated value is also a sunk cost: the fine-tuning investment cannot be recovered, and resulting model weights may not be portable to a competing platform's runtime environment, particularly where the base model is provider specific.

6.3.5. P5 - Within-User Learning and Compounding

On-demand inference pricing through Amazon Bedrock is the primary mechanism through which P5 operates in the AWS architecture. Every API call against Bedrock's foundation model access layer, every agent workflow built on Bedrock Agents, every knowledge base indexed through Bedrock Knowledge Bases, and every guardrail configured through Bedrock Guardrails incrementally deepens calibration to AWS's specific interfaces, authentication patterns, and response structures. The ratchet

effect embedded in AWS's commitment tier structures reinforces this at the financial level. Enterprises enabling provisioned throughput for peak workloads gradually normalise above-baseline consumption as teams calibrate workflows to performance guarantees. Each renewal at higher capacity reflects both genuine workload growth and operational dependency on performance characteristics. Swann (2009, pp. 84-85) notes that switching costs alone do not produce complete reluctance to move but raise the threshold. The ratchet effect raises that threshold incrementally with each renewal cycle, without any single decision identifiable as the moment of lock-in.

6.3.6. P6 - Persistence Despite Regulatory Intervention

The AWS Customer Agreement and AWS Service Terms (Amazon Web Services, 2026e; 2026i) make explicit what is contractually governed-data handling, service change notice periods, and acceptable use boundaries. Notably, AWS commits to providing at least twelve months' prior notice before discontinuing material service functionality and limits use of customer data to operating and maintaining services. These contractual protections define the visible and legally contestable surface, but the most durable switching costs are not in these terms. The AWS Responsible AI Policy (Amazon Web Services, 2026h) places compliance responsibility explicitly on the enterprise: enterprises are responsible for all decisions made and actions taken using AWS AI services and must ensure compliance with applicable laws including the EU AI Act. For enterprises in regulated sectors like financial services, healthcare, critical infrastructure, this means AWS's compliance infrastructure, audit tooling, CloudTrail logging, and IAM governance framework become operationally necessary for demonstrating compliance. The enterprise cannot straightforwardly migrate without reconstructing an equivalent compliance posture (audit trails, access logs, governance policies) in an entirely different technical environment. Regulatory frameworks targeting technical portability address only the data migration dimension. Opara-Martins et al. (2016) establish that cloud lock-in is a business risk extending far beyond the technical layer. In regulated sectors, the environment functions as an amplifier of lock-in rather than a constraint. An enterprise with mandatory data residency, government procurement, or sector-specific security requirements faces a binary choice. It can adopt AWS's premium GovCloud configuration, where the Free tier is unavailable, AWS IAM Identity Centre authentication is mandatory, and pricing carries a significant premium. Alternatively, it can undertake the substantial compliance burden of constructing an equivalent regulatory environment elsewhere. The compliance requirements that appear to constrain AWS's pricing power are the same requirements making AWS's compliance-certified infrastructure structurally necessary. This is the most durable switching cost form because it is created not by AWS's commercial decisions but by the enterprise's external regulatory obligations.

6.4. Cross-unit synthesis: Platform-generic and platform-specific mechanisms

Following Yin's (2014, pp. 143-152) pattern matching logic, this synthesis identifies which mechanisms are observable across both embedded units and which are specific to one.

6.4.1. Platform-Generic Mechanisms - Observable Across Both Units

Six switching cost mechanisms emerge consistently across Azure and AWS, suggesting structural characteristics of enterprise AI platform ecosystems rather than individual provider strategy.

Multi-layer cross-stack dependency activation. Both platforms manifest the vertical AI stack dynamic: dependencies at one layer activate and reinforce dependencies at adjacent layers, such that the total switching cost is a function of the full stack relationship rather than any single service within it. Azure AI Foundry simultaneously generates dependencies at the model access, identity, observability, and data storage layers; Amazon Bedrock generates parallel dependencies at the same layers in AWS's stack. Both demonstrate that AI integration activates multiple switching cost types simultaneously through the vertical AI stack rather than through any single layer in isolation. Hagiu

and Wright (2025) describe this as the structural consequence of vertical stack integration, where dominance at one layer is leveraged across others such that switching costs at any single layer are inseparable from those at every adjacent layer. Rong et al. (2026) explain the same dynamic from the platform strategy side as the all-in-one platform pattern: multiproduct ecosystems create interdependencies that make partial exit structurally irrational, because departure from any single component disrupts the value derived from the integrated whole. This supports P1 across both units.

Pricing abstraction as an information asymmetry mechanism. Both platforms use abstracted pricing units-Azure tokens, PTUs, and commitment credits; AWS tokens, TPUs, and Savings Plan hours-that decouple enterprise consumption from underlying compute and inference economics. These units do not map cleanly across providers, making cost comparison structurally difficult and discouraging multi-homing beyond list price effects. Consistent with He et al. (2023), such asymmetries magnify switching costs by weakening price transparency, echoing Farrell and Shapiro (1988) on mechanisms that raise the effective cost of alternatives via commitment or opacity. This supports P3. The OECD (2025) corroborates this, identifying lack of pricing transparency and standards as a structural feature reinforcing competitive barriers across AI infrastructure providers.

Commitment-based financial lock-in as a value/dependency trade-off. Both platforms embed their core value propositions like performance guarantees, cost predictability, and capability access within commitment structures that are financially punitive to exit. Azure's PTUs and EDPs and AWS's SageMaker Savings Plans and Bedrock TPU reservations follow the bilateral specificity logic identified by Besanko et al. (2017, pp. 364-365): the enterprise capturing the greatest operational value also incurs the highest financial switching cost. This reflects a platform-generic instantiation of P4 through structurally analogous but distinct commercial instruments across AI stack layers.

Marketplace integration as ecosystem-level switching cost amplification. Both Azure Marketplace and AWS Marketplace embed many third-party AI services (data labelling, monitoring platforms, compliance solutions, and domain-specific models) under commercial terms natively integrated with each platform's billing and identity governance frameworks (Microsoft, 2026d; Amazon Web Services, 2026g). Following Costabile (2024), this consolidation reflects ecosystem governance exercised through access rules, coordination mechanisms, and pricing integration that shape how firms interact with external complementors. As demonstrated by He et al. (2023), when switching costs interact with network externalities arising from a dense complementor ecosystem, their combined effect exceeds that of either mechanism operating in isolation. The cumulative effect across both platforms is the same: enterprises consolidating procurement, identity, and governance through marketplace channels make the value of the ecosystem itself non-transferable, producing a migration burden that operates above and beyond the switching costs associated with first-party services.

Partner programme stratification as ISV-side switching cost amplification. Both Azure and AWS operate stratified partner programmes where we find the Microsoft Cloud Partner Programme (Microsoft, 2026g) and the AWS Partner Network (Amazon Web Services, 2026g) tying ISV status to platform-specific certifications, integration patterns, and revenue-share structures. The resulting ISV dependencies extend ecosystem-level switching costs from the enterprise-platform relationship to the complementor-platform relationship: ISVs invested in platform-specific certification face switching costs in supporting equivalent functionality on competing platforms, limiting multi-cloud availability of complementary services. This extends the ecosystem governance perspective of Costabile (2024) and Chen et al. (2022) by identifying complementor-side switching costs as a structural amplifier of platform-side switching costs.

Regulatory compliance infrastructure as a lock-in amplifier. Both platforms provide compliance-certified infrastructure that regulated-sector enterprises come to depend on operationally. The within-unit analyses reveal that regulatory requirements do not constrain platform dependency but amplify it: the compliance posture built within a platform's governance infrastructure is not portable regardless of data portability regulations. This supports P6 and extends Opara-Martins et al. (2016) and the OECD (2025) finding that structural ecosystem characteristics shape competition. The UK Competition and Markets Authority's (2025) cloud services market investigation independently corroborates this

through primary data collection from both providers, finding that egress fees and technical barriers create adverse effects by *„locking customers into their initial choice of provider "*(CMA, 2025, para. 36b), while concluding that committed spend agreements in their current form do not harm competition (CMA, 2025, para. 28), empirically supporting the present study's argument that lock-in operates primarily through structural rather than contractual mechanisms.

6.4.2. Platform-Specific Mechanisms - Distinctive to Individual Units

Azure: Foundation model exclusivity as capability non-portability. Azure's lock-in at the model layer operates through exclusive or preferential access to OpenAI's GPT-4o and related foundation models under enterprise-grade managed service terms providing capabilities unavailable on competing platforms at equivalent terms. This creates a switching cost qualitatively distinct from financial commitment or pricing abstraction: capability non-portability, where the enterprise cannot simply pay to replicate the capability elsewhere. This reflects Microsoft's strategic OpenAI partnership as a structural source of competitive advantage. Evans and Schmalensee (2016, pp. 188-199) identify exclusive access as a defining characteristic of platform disruption, and this is the clearest instantiation of P2: the value-creating feature is the same feature making the capability non-portable.

AWS: Unified API aggregation as integration-layer lock-in. AWS's lock-in operates through the opposite logic: aggregation rather than exclusivity. Bedrock's value proposition is unified API access to multiple frontier models through a single authentication, billing, and security framework. The switching cost is not the inability to access a model elsewhere but the re-engineering cost of rebuilding the integration layer Bedrock's unified API provides. This reflects a deliberately open-but-integrated architectural philosophy whose stickiness operates through convenience and workflow encoding rather than capability exclusivity. The distinction between AWS's access-granting and Azure's tighter control over model access reflects the two platform-opening approaches identified by Boudreau (2010), who shows these produce qualitatively different innovation and dependency dynamics. AWS's ecosystem governance operates through provisioned throughput commitments, marketplace integration, and partner programme stratification rather than Microsoft's cross-product enterprise software licensing. The UK Competition and Markets Authority (2025) identify such cross-product effects as arising from Microsoft's licensing of Windows Server, SQL Server, and related productivity software (paras. 31, 33, 35), reflecting the different product portfolios of the two providers.

6.4.3. Within-User Learning as a Cross-Cutting Compounding Mechanism

Across both units, within-user learning effects continuously deepen switching costs through normal platform usage without deliberate action. In Azure, this operates through accumulated organisational knowledge around Azure ML Studio's pipeline conventions, AI Foundry's prompt flow configurations, and Microsoft Entra's identity framework. In AWS, through progressive calibration to SageMaker's SDK patterns, Bedrock's agent and knowledge base structures, and AWS IAM's access control conventions. Hagiu and Wright (2025) provide the theoretical grounding: as platform learns enterprise usage patterns and enterprise accumulates platform-specific institutional knowledge, the cost of moving to a platform lacking this mutual learning rises continuously. This supports P5 across both units and constitutes the conceptual framework's feedback loop: switching costs compound autonomously regardless of which specific mechanism originally generated them.

6.5. Case-Level findings: RQ1

The cross-unit synthesis yields the following case-level findings in response to RQ1.

Enterprise AI infrastructure platforms generate switching costs through a layered architecture of dependency operating simultaneously across three dimensions. First, at the governance layer, pricing

abstraction and commitment structures create financial and informational barriers embedded in the platform's value proposition rather than in explicitly restrictive mechanisms - supporting P2 and P3. Second, at the technical and organisational layer, proprietary APIs, managed service toolchains, and identity integration create workflow and capability dependencies compounding through within-user learning - supporting P1 and P5. Third, at the institutional layer, compliance infrastructure and regulatory requirement alignment create dependencies structurally durable and amplified by the regulatory frameworks designed to constrain them - supporting P6. Cutting across all three is the P4 trade-off: features delivering the highest enterprise value at each layer are the same features generating the highest switching costs. The most analytically significant supply-side finding is the empirical characterisation of how structural inseparability, a dynamic theorised in established literature but not previously operationalised across multiple simultaneous layers and which manifests distinctively across both platforms. Whether through Azure's foundation model exclusivity or AWS's unified API aggregation, mechanisms generating the highest value are the same generating the highest switching costs, through structurally different pathways producing functionally equivalent outcomes. This supports P4 at the case level and extends classical switching cost theory by demonstrating how this dynamic operates across the interdependent layers of a vertically integrated AI stack, a multi-layer configuration the canonical single-layer formulations do not directly anticipate. This carries direct implications for enterprise governance and regulatory design. These supply-side findings are integrated with demand-side evidence in the subsequent chapter, where enterprise strategic responses are examined in relation to P4, P5, and P6.

PART TWO: DEMAND-SIDE WITHIN-UNIT ANALYSIS (PILOT INTERVIEWS)

6.6. Introduction to Demand-Side Analysis

The demand-side analysis examines how enterprise organisations experience, interpret, and strategically respond to the switching cost mechanisms identified in the supply-side analysis of Microsoft Azure and AWS. This constitutes the Type B embedded unit analysis within the study's Type 2 embedded single-case design, addressing RQ2: How do enterprise firms navigate switching cost exposure in enterprise AI infrastructure platforms?

Data for this analysis were drawn from four pilot interviews conducted in April-May 2026. The sample spans three industry sectors: telecommunications (two informants), federal services / management consulting (one informant), and financial services (one informant). Role types cover Vice President of AI Transformation (R&D), Head of AI Transformation, Senior Enterprise Solutions Architect, and Head of AI in a regulated banking environment. Interview duration ranged from approximately 45 to 65 minutes, and informant experience ranged from 15 to 25 or more years. In accordance with the agreed confidentiality protocol, informants are identified throughout the analysis by the codes shown in Table 5. No personal names, organisation names, or vendor-specific product names are retained where their inclusion would compromise anonymity.

It is important to note the methodological status of this analysis. The demand-side interviews are pilot interviews - exploratory in design and purpose. They are not intended to confirm or statistically generalise the demand-side propositions. Rather, they generate preliminary demand-side evidence that supports analytical generalisation within the single overarching case. Findings from this section are therefore framed as patterns observed and propositions developed, consistent with the exploratory framing appropriate to a small purposive sample. Informant accounts were coded using the A-G-N attribution framework established earlier in the study. The analysis proceeds in three sections. Section 6.7 examines how enterprise informants experienced and characterised the supply-side mechanisms identified in Part 1, providing demand-side triangulation of P1-P3. Section 6.8 examines the strategic responses enterprises adopted to manage switching cost exposure, directly addressing RQ2 through P4-P6. Section 6.9 synthesises findings across informants to identify common and divergent patterns.

6.7. How enterprises experience supply-side mechanisms

This section attempts to provide demand-side evidence for P1-P3, triangulating the supply-side documentary analysis with informant accounts of how switching cost mechanisms are experienced in practice. Where informant accounts converge with the supply-side analysis, construct validity is strengthened through independent source convergence. Where they diverge, the divergence is examined for whether it extends, qualifies, or challenges the relevant proposition.

6.7.1. P1 - multi-layer dependency as experienced by enterprise informants

The supply-side analysis established that AI platform integration activates switching costs simultaneously across multiple stack layers. Pilot interviews support this finding: each of the four informants described switching cost experiences extending well beyond a single stack layer.

Architect-FedSvc described an environment with cross-layer simultaneous dependencies: *"We had technical barriers, cultural barriers, contractual barriers, and even organisational ones. We had to rewrite pipelines, retrain models, and reconfigure integrations"* **(Architect-FedSvc, A/G).** Head-Telco's characterisation describes the co-existence of contractual, ecosystem and organisational dependencies as distinct layers contributing to the overall switching cost: *"If anything, it will be a switch to a different company or a different solution, but not to not using it. The contract is the biggest dependency right now. But there are also ongoing changes in the organisational structure, and the ecosystem around the solution creates a lock-in effect."* **(Head-Telco, G/A).** VP-Telco's account distinguishes two dimensions of dependency - technical architecture and organisational process while noting that the organisational dimension is harder to quantify and is substantially larger in practice: *"The technical switching cost - replacing APIs, re-implementing integrations, retraining models is significant but estimable. The organisational switching cost - retraining, productivity loss, change management is harder to quantify and in my view substantially larger. "(***VP-Telco, A/N).** Head-FinSvc introduced a qualitatively distinct variant of multi-layer dependency that was not prominent in the prior three accounts: the pre-regulatory lock-in layer. He described how data sovereignty regulation (Swiss FINMA requirements and EU data residency rules) imposed a binding constraint on which cloud regions, and therefore which AI capabilities were permissible before any architectural or commercial decision was made. This regulatory pre-constrain created a primary dependency layer that operates upstream of the technical and contractual layers also described by the other informants: "*We have to leverage the cloud here in our country. We can't use the EU or other areas for the data we're hosting. everything around our data needs to be processed and stored in our country.*" **(Head-FinSvc, G)**

This account supports P1's multi-layer dependency finding and extension: in heavily regulated sectors, a regulatory compliance layer sits above and constrains the technical and commercial layers described in the supply-side analysis. The most consistently reported unexpected layer across all four informants was the organizational and human capital dimension - a layer not foregrounded in the supply-side analysis and identified as an emergent finding in Section 7.2.

6.7.2. P2 - Value-Embedded Switching Costs as Experienced by Informants

P2 proposes that the most significant switching costs are embedded in value-creating features rather than deliberately restrictive mechanisms. Pilot interview evidence supports this proposition across the four informants, with each describing a pattern in which the capabilities delivering highest operational value are the same capabilities generating deepest lock-in.

*"The pattern is clear. The features we value most are the ones creating the deepest lock-in. I do not think this is accidental on the part of the vendor. "(***VP-Telco, A/N).** *"We have seen gains up to lead time reduction of twenty-five, fifty percent. So, it would be very hard to remove it entirely. But it is*

*always a cost thing - if the AI is more expensive than a human, we need to rethink. "(***Head-Telco, G/N)**. **Architect-FedSvc** described the same dynamic in enterprise-scale terms *"Ironically and unfortunately, the most valuable capabilities are also the ones that create the deepest dependencies. The more operationally critical and differentiated the capability, the deeper the lock-in. ""We would love to have Claude Anthropic 4.6 or whatever in-house, but it's exactly the challenge that we have. In the Azure Cloud you could actually use it, but then the data processing will be outside of the country"* **(Head-FinSvc, G)**

This account supports P2 in an analytically significant way. Where P2 frames value-embedded switching costs as architecturally designed by the platform, Head-FinSvc's experience shows that the same trade-off can arise from regulatory constraints rather than platform design. The value is present and acknowledged, but access is blocked by a compliance layer the enterprise cannot alter. The practical effect at the enterprise level is identical, but the causal mechanism differs in regulatory amplification of value-embedded dependency rather than platform-designed architecture. Each informant described the decision to accept lock-in as a conscious strategic choice rather than discovering it retrospectively, which is a qualification developed in the integrated finding for P2.

6.7.3. P3 - Ecosystem Governance Mechanisms as Experienced by Informants

P3 proposes that ecosystem governance mechanisms generate switching costs at the ecosystem level rather than the individual service level. Pilot interview evidence lends support to this proposition, with contractual commitment structures and vendor ecosystem dependencies surfacing as primary governance-layer mechanisms. *"We committed to one supplier, and that was part of the deal. There is an ecosystem around the solution - concepts like agents and skills and such - which creates a lock-in effect for sure. "(***Head-Telco, G)**. *"We evaluated a proprietary orchestration capability that was impressive - genuinely better than the open-source alternative at that moment. But the skills, the tooling, and the workflow concepts were entirely provider specific. An engineer trained on it would have difficulty transferring those skills to a different environment". (***VP-Telco, N/G)**. *"These dependencies are heavily tied to provider-specific AI systems. The provider-specific APIs, SDKs, and managed services are all tightly integrated. The most significant barrier in practice was technical complexity, especially when AI is embedded across multiple layers of the stack. ". (***Architect-FedSvc, A/G)**. *"Making separation from the Microsoft stack progressively more costly. So of course, there are advantages when you use Microsoft. I think also how they connect all the different parts, also the guardrail piece as an example. It's great in one way, another way it's also painful"* **(Head-FinSvc, G/N)**.

Pricing opacity was referenced by Head-Telco and Head-FinSvc in terms of commitment tier structures (volume discounts, consumption pricing) that create disincentives to exit. Head-FinSvc described the dynamic explicitly when asked whether a price increase would lead to migration: the answer was framed as case-by-case, with the assessment driven by the cost of maintaining dual infrastructure rather than by the price change alone.

6.8. Enterprise strategic management of switching costs

This section presents the primary RQ2 analysis, examining how enterprise firms navigate the trade-off between platform value capture and switching cost exposure. Given the pilot nature of this sample, findings are framed as observed patterns and tentative propositions rather than definitive conclusions about enterprise behaviour generally. Three strategic dimensions are examined: architectural strategies, platform management strategies, and governance and contracting strategies.

6.8.1. Architectural Strategies

Architectural strategies address switching cost exposure through technical design choices - abstraction layers, modular architectures, open-standard tooling, and portability-by-design approaches. Pilot interview evidence suggests deliberate architectural mitigation is present but unevenly distributed across the sample. VP-Telco offered two concrete examples of architectural choices made to preserve flexibility at the cost of some capability premium: *"We chose an open-standard monitoring approach that was perhaps fifty percent of the capability but preserved our architectural flexibility. That decision was made formally, through our enterprise architecture governance process.* ***"(*VP-Telco, A)**. *"We decided not to go directly and start working on the vendors' AI platforms, but to use those platforms to build our own custom AI integration tools. The strategy is to slowly but surely get our capabilities out of their systems. "(***Architect-FedSvc, A)**. *"We are trying to build technical guardrails - filtering mechanisms to anonymise data and to put up a more controlled data sharing model - common practices like MCPs and vector databases. "(***Head-Telco, A/G)**. *"From the beginning, we saw how quickly things are developing. So therefore, we were saying we go slow but steady. And we want to have architecture which is really open and flexible"* ***(Head-FinSvc, A)***

This account introduces a capability-tier distinction not prominent in the prior three accounts: architectural portability is not homogeneous across use cases. For lightweight productivity tasks, portability is high and switching costs are low; for deeply integrated operational workflows, switching costs persist even within a modular architecture. This tiered portability finding represents a qualification of architectural strategy literature that deserves further empirical investigation.

6.8.2. Platform Management Strategies

Platform management strategies include multi-cloud diversification, selective adoption of proprietary services, and the deliberate allocation of workloads across providers to maintain competitive leverage. Pilot interview evidence suggests multi-provider approaches take different forms with informants' responses ranging from operational multi-cloud (Architect-FedSvc) through single-provider with a backup cloud (Head-FinSvc) to single-provider with aspirational diversification (Head-Telco).

"It may sound compelling at a broad level, but in practice it is very complex and costly. The workloads are segmented - each provider owns a distinct domain. We do not mix them presently, but our end goal is to eventually get the best of it." **(Architect-FedSvc, G/A)**. *"That relationship is deliberately bounded. We avoid their proprietary data formats where we can. The integration is functional but not structural."* **(VP-Telco, A/G)**. *"In the long run it would be very beneficial if we could have a couple of vendors handled in a safe manner with data governance in place. But that is not the case today - we committed to one supplier as part of the deal. "(***Head-Telco, G)**. "*We have on-prem and we have a cloud provider. I think it's a good combination, yes, and we do have a backup cloud, so with another provider."* **(Head-FinSvc, G).**

Multi-provider strategy was used as negotiating leverage by Architect-FedSvc (exit-strategy blueprints as a credible-alternatives signal) and by VP-Telco (open-source capability cultivation as a strategic hedge). Head-FinSvc's account suggests that in regulated sectors, faced with hyperscalers, negotiating leverage through multi-provider credibility is structurally limited by market position.

6.8.3. Governance and Contracting Strategies

Governance and contracting strategies include formal AI vendor governance frameworks, exit planning, contract negotiation for portability and termination and regulatory leverage in platform negotiations. *"The exit plan has three tiers based on the triggering scenario: pricing escalation, capability deprecation, and service disruption. "****(VP-Telco, G)****. "Just the mere fact that exit strategies exist, and the vendors know they exist, strengthens our negotiating position. We have a work*

*stream focused only on exit strategy - to put us in a position that vendors know we are preparing to exit. "(***Architect-FedSvc, G).** *"Of course we have an exit. But I am not handling the contracts. I have not been in those negotiations" (***Head-Telco, G)**. *"We do have an exit strategy on the cloud. We have to have them in place. That's been, in our case, it's by law" **(Head-FinSvc, G)**.

The VP-Telco and Architect-FedSvc described governance maturity developing in response to accumulated dependency, and Head-Telco described relatively informal governance despite deep integration.

However, Head-FinSvc presented a different pattern entirely. Governance formality was not built in response to accumulated dependency but was mandated by Swiss FINMA regulation before significant AI integration had even begun. Exit plans, risk classifications, and resilience testing are legal obligations, not lessons learned from experience. This offers an important qualification: governance sophistication lags integration broadly but breaks down in heavily regulated sectors like finance where regulation forces governance discipline from the outset, independent of how deep the dependency actually is.

6.8.4. P4 - The Trade-off as Lived Enterprise Experience

P4 proposes that enterprise strategies for managing switching costs involve a trade-off between platform value capture and lock-in exposure, where the features delivering the greatest operational value typically generate the highest switching costs. Pilot evidence from all four informants is consistent with this trade-off as a lived experience, but the four accounts collectively reveal that the P4 trade-off is navigated through qualitatively different mechanisms depending on the enterprise's regulatory context and integration maturity. *"The decision to integrate is deliberate. The workflows become more dependent. I think it is closer to exponential in the early phases."* **(VP-Telco, N/A)**.

*"Early in our adoption, we had very strong leverage - we had options across vendors. But over the last three years, as AI has become embedded in mission-critical systems, providers are gaining leverage, especially around pricing and contract terms. "***(Architect-FedSvc, G).**

"*The whole thing is a transformation. What we don't want to do is that we change a model and people get frustrated because the results have changed and maybe to the worse*" **(Head-FinSvc, N/G)**

This supports P4 across all four informants and adds an important dimension: the value/lock-in trade-off is not merely an abstract strategic calculation but a lived organizational experience whose salience is shaped by regulatory context, competitive pressure, and user adoption dynamics. Together, the four accounts suggest that P4 is a structurally robust finding but that the mechanism through which enterprises navigate it varies systematically with context. (Section 6.10).

6.8.5. P5 - Within-User Learning as Lived Enterprise Experience

P5 proposes that within-user learning effects cause switching costs to compound autonomously through normal platform usage, such that lock-in deepens over time without deliberate action by either the platform or the enterprise. Pilot interview evidence from all four informants supports this proposition, with Head-FinSvc adding a distinctive dimension: within-user learning effects manifest differently in regulated environments where adoption is constrained to mature or approved capabilities. *"The people are using all these AI tools also privately. So, they are using different models anyway. And they also compared in our internal secure version with the AI versions that they use on their private devices."* **(Head-FinSvc, N).**

*"This is not linear - I think it is closer to exponential in the early phases." (***VP-Telco, N).**

*"I don't think anybody wants to go back. It has improved efficiency already." (***Head-Telco, N).**

*"When skills, training and operational habits become aligned with a specific platform, it makes it even harder to switch and more difficult than the technical migration itself. "(***Architect-FedSvc, N).**

This account lends support to P5 in a specific context: where users maintain simultaneous exposure to multiple AI interfaces, including consumer-grade tools with broader capability sets. The depth of within-user learning lock-in to a single enterprise platform may be attenuated. However, Head-FinSvc simultaneously noted that users actively complain about the capability gap between the internal enterprise platform and the consumer tools they use privately, creating a reverse pressure: not that users are locked into the enterprise platform, but that the enterprise platform is perceived as inferior relative to the consumer baseline. This might drive demand for more capable (and potentially more lock-in-inducing) enterprise models. The net effect on P5 is a qualification rather than a challenge: within-user learning compounding operates more strongly at the use-case-specific layer than at the general productivity layer in organizations where consumer AI exposure is widespread.

6.8.6. P6 - Regulatory Amplification as Lived Enterprise Experience

P6 proposes that structural switching costs in AI-integrated cloud ecosystems persist despite regulatory intervention mandating portability and interoperability, and that in practice, compliance requirements amplify rather than constrain platform dependency. Pilot interview evidence from all four informants supports this proposition. Head-FinSvc provides the most direct and analytically complete account of regulatory amplification, because data sovereignty requirements constitute the primary constraint on their entire AI platform strategy rather than a secondary consideration. The other three informant described regulatory amplification as an emergent effect: compliance infrastructure built on provider-native tooling became a source of dependency that regulatory portability mandates did not address: *"Regulation encourages data portability, auditability, and transparency in theory. But in practice it increases lock-in, because you are going to depend on the provider's tooling to be able to do all of that. Compliance is tightly coupled with the provider ecosystem. That is the paradox we are facing." (***Architect-FedSvc, G).**

*"The models we use are all from the US, but they are executed in the EU. Where the actual data resides is the most important thing. We believe we have zero data handling outside of the EU. But the data governance model is definitely something we are still shaping up. "(***Head-Telco, G).**

*"The compliance infrastructure itself has become a dependency. Our audit trail systems, our model risk management tooling, our explainability frameworks. But the practical effect at the implementation layer is to create additional stickiness, not less." (***VP-Telco, G/A).**

Head-FinSvc's also extends P6 by demonstrating that regulatory amplification can operate as a structural constraint that precedes AI adoption rather than emerging from it. *"Our data residency requirements for banking data - which mandate that all data processing occurs within our infrastructure - effectively eliminate competitive alternatives to Microsoft Azure's in-region for LLM hosting, creating a regulatory-induced monopoly on permissible providers…We don't have the latest and the greatest technology in our cloud here in our country. That's one of the policies that we have"* **(Head-FinSvc, G)**

This account extends the regulatory paradox identified across the other three prior informants. Where Architect-FedSvc and VP-Telco ,Head-Telco described compliance implementations that deepened existing provider dependencies, Head-FinSvc describes a context in which regulation eliminates provider choice before any dependency is formed which signals a stronger form of regulatory amplification in which the regulatory framework functions as an entry barrier to alternative providers rather than merely as an amplifier of existing switching costs. The implication is that P6 may operate

through two distinct mechanisms depending on the regulatory context: as a post-adoption amplifier (the compliance paradox observed in EU and US regulatory environments) and as a pre-adoption constraint (the provider selection barrier observed in Swiss banking regulation). Both produce the same outcome of deepened provider dependency but through structurally different causal pathways.

6.9. Demand-Side Cross-Informant Synthesis

This section synthesises patterns across the four pilot informants, identifying common patterns, notable divergences, and sector-specific observations. Given the pilot nature of the sample, these are presented as tentative analytical patterns rather than statistically generalisable findings.

The most consistent finding across all four informants was the characterization of switching costs as multi-dimensional and compounding rather than bounded and stable. Each described accumulation as ongoing and self-reinforcing through normal usage (consistent with P5), and three of the four informants (VP-Telco, Architect-FedSvc, and Head-FinSvc) explicitly identified the organizational and human capital dimension - skill development, workflow habituation, user adoption - as the most underestimated component of total switching cost. This pattern was observable across all three sectors and all four role types, suggesting it may be a robust demand-side finding rather than a sector-specific artefact. A second consistent pattern was the coexistence of strategic awareness of lock-in risk with continued deepening of dependency. None of the four informants described their organization as unaware of vendor concentration risk; all described active governance responses. Yet all also described continued deepening engagement with primary vendors, suggesting a pattern of strategic acceptance - where enterprises accept dependency depth as the cost of capability access - rather than strategic naivety.

"We have seen gains up to lead time reduction of twenty-five to fifty percent. It would be very hard to remove AI entirely. But the right mantra is to be agile." (Head-Telco, N/G).

A third but most important pattern observed with Head-FinSvc, is that mission-criticality and not integration depth determines where enterprises invest in governance. All four informants described a two-speed posture: mission-critical workflows attract formal exit planning, resilience testing, and fallback alternatives and non-mission-critical AI applications often involve accepted levels of provider dependency with limited oversight, fallback alternatives, or formal exit planning. Head-FinSvc illustrated this directly saying fraud detection always has a backup because payments cannot stop, but clients onboarding without AI for a day are simply accepted. This qualifies the governance-lag finding from the prior three-informant analysis: governance formality does lag integration depth in general but converges rapidly for mission-critical workloads regardless of sector. The implication is that enterprises are not uniformly slow to govern AI dependency - they are selectively disciplined, and the trigger is operational consequence and not dependency depth alone.

PART THREE: INTEGRATED CASE-LEVEL SYNTHESIS

6.10. Integration of Supply-Side and Demand-Side Findings

This section performs the integrated case-level synthesis, bringing together the supply-side within-unit analyses of Microsoft Azure and Amazon Web Services with the demand-side pilot interview analysis. Following Yin's explanation building logic, findings from both embedded unit types are compared against the six theoretical propositions to construct an integrated account of the phenomenon.

P1 - Integrated finding: Integrating the supply-side analysis with the pilot evidence yields that AI integration creates switching costs across multiple platform layers simultaneously, and that these layers interact, making the total cost of switching higher than any single layer would suggest. The demand-side evidence supports this and adds two layers the supply-side framework did not capture.

The first is the human capital. All four informants independently identified retraining staff, rebuilding workflows, and managing organizational change as the most underestimated part of switching costs, harder to quantify than technical migration, but often more costly in practice. The second is the pre-regulatory constraint layer, conveyed by Head-FinSvc. "*In our banking, data residency law determines which cloud providers are even permissible before any technical or commercial decision is made*". Regulation does not amplify dependency here - it creates it from the outset by eliminating alternatives before integration begins. The integrated finding is therefore that switching costs in AI platforms are not just technical or contractual but are also human and regulatory. A complete picture must include all four layers: technical, governance, human capital, and in regulated sectors, pre-regulatory provider constraint.

P2 - Integrated finding: The supply-side analysis established that the deepest switching costs are embedded in value-creating features rather than in deliberately restrictive mechanisms such as egress fees. The demand-side evidence adds two qualifications. First, enterprises accept lock-in consciously because the value justifies it; what they underestimate is how fast dependency compounds through normal usage, suggesting P2 and P5 are the same mechanism seen from different angles. Second, Head-FinSvc shows that value-embedded dependency does not always originate from platform design - Swiss data residency law locks the enterprise into Azure by regulation rather than architecture, producing an outcome identical to P2 from a different cause. The integrated finding is that value-embedded switching costs arise from three intersecting forces like platform architecture, enterprise integration choices, and regulatory constraints with the dominant force varying by sector.

P3 - Integrated finding: The supply-side analysis found that pricing structures, commitment agreements, and marketplace dependencies create switching costs at the ecosystem level rather than at the level of individual services. The demand-side evidence adds two dimensions. First, a temporal one: all four informants described how negotiating leverage erodes progressively as commitment lengthens, so what felt like a fair deal at signing becomes increasingly difficult to renegotiate. Second, Head-FinSvc shows the relevant ecosystem is wider than AI alone - with Microsoft's Office 365, SharePoint, and desktop environment already embedded in daily work and AI now layered on top, separating from Microsoft is far more costly than an AI-only assessment would suggest. The integrated finding is that ecosystem-level switching costs compound over time and extend well beyond the AI stack itself. A rival account that enterprises can neutralise ecosystem governance through selective multi-homing and open-source alternatives is not supported at production scale, where provisioned throughput non-transferability makes parallel capacity economically prohibitive.

P4 - Integrated finding: The supply-side analysis found that the value/lock-in trade-off is built into platform architecture. All four informants support this but navigate it differently. VP-Telco and Architect-FedSvc accepted lock-in consciously. Head-Telco was driven by competitive pressure to adopt the most capable model regardless of lock-in risk, and Head-FinSvc had no choice because regulation pre-selected the only permissible provider. The integrated finding is that the trade-off is not a one-time decision but a compounding process: every period of value extraction deepens the cost of future exit. The mechanism varies by context - strategic, competitive, or regulatory - but the outcome is the same: dependency grows faster than governance frameworks anticipate, and point-in-time assessments systematically understate forward exposure. An alternative explanation that enterprises with better governance could avoid the trade-off entirely is only partially supported; stronger governance reduces dependency exposure but does not eliminate the trade-off as all four informants experienced deepening lock-in regardless of governance maturity.

P5 - Integrated finding: The supply-side analysis identified within-user learning through model customization, prompt caching, and accumulated system prompts as a self-deepening lock-in mechanism. The demand-side evidence confirms the human capital layer compounds it further. As staff build platform-specific skills, habits, and workflows, the cost of switching becomes less about migrating technical assets and more about retraining people - which all four informants described as

harder to reverse and more disruptive than the technical migration itself. P5 is therefore partially supported rather than fully confirmed. Within-user learning locks enterprises across both the technical and human capital layers, but its strength varies by use-case depth and is offset by a rival dynamic: where users maintain broad exposure to competing AI tools outside the enterprise, that exposure can reduce single-platform stickiness rather than deepen it. The autonomous-compounding claim in P5's strong form is thus supported only at the use-case-specific layer, and this disconfirming evidence is retained rather than set aside.

P6 - Integrated finding: The supply-side analysis identified compliance infrastructure as a lock-in amplifier in regulated sectors. Demand-side analysis supports this across all three sectors and further reveals a regulatory paradox: compliance frameworks intended to support portability can instead reinforce switching costs. This occurs through two mechanisms depending on when regulation intervenes relative to AI adoption. Post-adoption regulation, observed in telecommunications and federal services. Pre-adoption regulation, identified in financial services, constrains provider choice before integration occurs. Although both mechanisms deepen provider dependency, they operate through distinct pathways with different implications for regulatory design. The paradox appeared across two jurisdictions (US and EU/Swiss) and three regulatory regimes, indicating a broader cross-sectoral pattern rather than a sector-specific effect. P6 is therefore partially supported and time-bound rather than unconditionally confirmed: the evidence supports short-term amplification, but a rival account, in which portability mandates such as the EU Digital Markets Act erode these switching costs over the medium term, remains compatible with point-in-time data and is retained as a live alternative. Within that boundary, the integrated P6 finding strengthens and extends the supply-side analysis.

6.11. Case-Level Findings: RQ1 and RQ2

RQ1: What architectural design choices and ecosystem governance mechanisms create switching costs in enterprise AI infrastructure platforms?

The integrated supply-side and demand-side evidence supports the following case-level answer to RQ1. Enterprise AI infrastructure platforms generate switching costs through a layered architecture of dependency operating simultaneously across four analytically distinct dimensions, each confirmed by both supply-side documentary analysis and supported by demand-side pilot informant accounts. At the governance layer, pricing abstraction and long-term commitment structures create financial and informational barriers to exit. These are embedded in the platform's value proposition rather than in explicitly restrictive mechanisms. The pattern is observable across both supply-side units and supported by each of the four demand-side accounts through references to single-supplier commitments, declining negotiating leverage, and ecosystem-level contractual obligations (P2, P3). At the technical and operational layer, proprietary APIs, managed service toolchains, and AI pipeline architectures create workflow and capability dependencies that compound through within-user learning. This was confirmed in the supply-side analysis through model fine-tuning and system prompt accumulation, and corroborated in the demand-side accounts, which describe multi-tier technical complexity and the long timelines required for migration (P1, P5).At the institutional layer, compliance infrastructure and regulatory requirement alignment create dependencies that are structurally durable and amplified by the regulatory frameworks intended to constrain them - supported by supply-side documentary analysis and by demand-side accounts from each of the four informants across three regulatory jurisdictions (P6). At the human capital layer - an analytically significant finding that extends the supply-side framework - skill specialisation and organisational knowledge accumulation create switching costs that may exceed technical architecture switching costs in practical significance and are largely invisible to standard TCE-based switching cost analysis (extension of P1 and P5). The most analytically significant integrated finding for RQ1 is the structural inseparability of value creation and dependency generation observable across both supply-side units and supported by each of the four demand-side accounts. The mechanisms generating the highest enterprise value such as fine-tuned models, integrated development environments, managed

compliance infrastructure - are architecturally identical to the mechanisms generating the highest switching costs. Two rival explanations were tested against this RQ1 account before it was accepted. The first concerns P1's multi-layer claim. A plausible alternative is that switching costs are concentrated at a single dominant layer, and that observed multi-layer dependency reflects enterprise architectural choices rather than a structural property of the platforms. The documentary and informant evidence does not support this concentration rival: dependencies were evidenced simultaneously at the commercial, model, data, platform, and human-capital layers, and migration timelines reflected cumulative rather than single-layer cost. The evidence does qualify P1 in one respect, however. Enterprises that adopt abstraction layers, multi-cloud configurations, or portability-by-design, proactively in the case of Head-FinSvc or reactively for VP-Telco and Architect-FedSvc reduce the depth of layers, so the magnitude of multi-layer lock-in is partly a decision variable operating within a structurally multi-layered architecture.

The second rival concerns P2's claim that the deepest switching costs are embedded in value-creating features rather than in deliberately restrictive mechanisms. The competing explanation is that restrictive mechanisms like egress fees, committed-spend lock-ins, and proprietary formats are the primary drivers. The evidence favours P2 over this rival: informants identified managed AI services, fine-tuned models, and integrated toolchains as the deepest sources of lock-in, while regulatory analysis (CMA, 2025) found egress fees and technical barriers to cause genuine but secondary harm and did not find current committed-spend agreements decisive. The rival is therefore not eliminated but subordinated: deliberately restrictive mechanisms contribute as a secondary layer, while the dominant and most durable costs remain those embedded in value creation, consistent with P2 and with the study's central finding of structural inseparability.

RQ2: How do enterprise firms navigate switching cost exposure in enterprise AI infrastructure platforms?

Pilot interview evidence suggests three tentative patterns in enterprise responses to switching costs, presented here as preliminary observations that warrant further empirical validation with a larger sample. First, firms manage dependency through architectural mitigation strategies ranging from proactive portability-by-design approaches established before adoption (Head-FinSvc) to reactive abstraction-layer investments introduced after dependency accumulation (VP-Telco; Architect-FedSvc).

Regulatory context appears significant: strongly regulated sectors show more proactive mitigation, while portability also varies by use case, remaining feasible for lightweight productivity tools but limited for deeply integrated workflows. Second, multi-provider strategies appear across the sample but function mainly as negotiation or resilience mechanisms rather than true operational interchangeability. Workloads remain segmented by provider, and viable multi-provider leverage depends heavily on enterprise scale and regulatory constraints. Head-FinSvc further identifies a boundary condition where data residency rules eliminate alternative providers entirely for some SMEs. Third, governance and exit planning arise through different mechanisms: reactively after dependency formation in telecommunications and federal services, but ex ante through mandatory resilience regulation in banking. Across all cases, enterprises ultimately accept the value-lock-in trade-off, while switching costs compound over time and progressively reduce strategic flexibility beyond what existing governance frameworks mechanisms often fail to mitigate effectively.

7. Discussion

7.1. Introduction

This chapter interprets the findings in relation to the research questions and theoretical framework. It restates and answers each research question, examines theoretical implications for switching cost theory, platform economics, and industrial economics in AI-integrated platform contexts, compares findings with previous research, draws practical implications, acknowledges limitations, and identifies directions for future research.

7.2. Conclusion

In answer to RQ1, the documentary analysis characterizes six interrelated switching cost mechanisms operating across the enterprise AI stack: cross-layer technical dependencies, ecosystem-level pricing and bundling structures, ecosystem governance, value-generating features inseparable from dependency creation, within-user learning effects, and compliance infrastructure dependencies. These mechanisms are inferred from vendors' published pricing, contractual, and technical documentation rather than observed against a counterfactual migration, and their validity rests on convergence across document types and corroboration from the demand-side accounts. Three of the six proved platform-generic, recurring across both AWS and Azure, while the platforms' distinctive architectural strategies like Azure's foundation model exclusivity and AWS's unified API aggregation represent contrasting routes to functionally equivalent switching costs rather than additional mechanisms. The most analytically significant supply-side finding is the empirical characterizations of structural inseparability - a dynamic theorized in the established platform-economics literature but not previously operationalized across multiple simultaneous layers as it manifests in the vertical AI stack, where a single provider's control across infrastructure, models, and applications produces cross-layer lock-in that classical single-layer formulations do not directly anticipate.

In answer to RQ2, three preliminary patterns emerge from the pilot evidence. Enterprises navigate the value/lock-in trade-off through strategic acceptance rather than avoidance (P4): they knowingly accept the trade-off at adoption, after which switching costs compound through normal usage and progressively narrow strategic flexibility. Within-user learning tends to deepen these costs at the use-case layer, but the effect is partially offset where users retain broad exposure to competing tools, so P5 holds only in qualified form. Structural switching costs persist despite regulatory intervention, because they are embedded in technical architecture and organizational processes rather than in the contractual terms that portability and interoperability remedies address; P6 is supported, with a contractual-primacy rival tested and subordinated rather than eliminated. These patterns carry one practical implication: switching cost governance cannot be treated as a post-integration optimization problem. Architectural portability decisions, governance frameworks, and exit planning are most consequential at initial adoption, before integration depth and organizational learning create compounding dependency. Multi-provider configurations appear across the sample but function primarily as negotiating and resilience instruments. The regulatory context appears to shape enterprise capacity to manage dependency. Strong regulated informants adopting more proactive governance than those whose governance developed reactively.

Two emergent findings extend the framework beyond the propositions. First, human capital dependency (skill specialization, workflow habituation, and organizational learning) may exceed technical switching costs in practical significance while remaining underexplored in transaction-cost-economics analysis. Second, the regulatory dynamic recurred in each sampled sector and across both regulatory jurisdictions, suggesting it may reflect a structural feature of enterprise AI ecosystems rather than a sector-specific phenomenon. Both are candidate patterns rather than established results.

These findings are exploratory; they support analytical generalization to the theoretical propositions but do not establish saturated empirical claims about enterprise behaviour and are best understood as candidate patterns warranting further empirical validation. validation.

7.3. Theoretical contributions and practical implications

This study contributes to switching cost theory, platform economics, and industrial economics in four ways. The first three contributions are grounded in the supply-side documentary analysis and supported by multiple independent sources; the fourth introduces a temporal dimension partially observed through the pilot interview sample and is therefore framed as candidate, warranting further empirical validation.

Strong contributions (supply-side grounded). First, the study extends classical switching cost theory (Klemperer, 1995; Farrell and Klemperer, 2007) to a setting where switching costs operate across interdependent ecosystem layers. The vertical AI stack identified by Hagiu and Wright (2025) creates a layered dependency structure in which switching costs at any single layer are inseparable from those at adjacent layers - a stacked configuration the classical theory does not directly anticipate. The study further shows that the financial-procedural-relational typology of Burnham, Frels and Mahajan (2003), developed in B2C consumer settings, generalises to enterprise AI contexts, with each cost type taking platform-specific forms. Second, the findings advance platform economics by demonstrating that lock-in is embedded within value creation rather than imposed through contractual restriction. This extends the bilateral specificity logic of Besanko et al. (2017, pp. 364 -365) to the ecosystem level identified by Jacobides et al. (2018) and Rong et al. (2026), in ways that exceed the dyadic supplier-buyer specificity originally theorised. Third, the study contributes to industrial economics by characterising a contemporary mechanism of incumbent advantage. Three platform-generic mechanisms recur across both AWS and Azure: pricing abstraction, value-embedded lock-in, and regulatory amplification. Together they point to a market likely to stay concentrated, because new entry alone is unlikely to erode them. This extends Besanko et al.'s (2017) concept of bilateral specificity to the multi-sided platform setting - the dependency here is co-created, arising from value-creating features the enterprise chooses to adopt rather than from transaction-specific investment alone.

Candidate contribution (demand-side, exploratory). Fourth, the study introduces a candidate dynamic perspective on switching cost accumulation, suggesting that within-user learning effects and platform-specific organisational know-how may combine to deepen lock-in over time without deliberate action. While the supply-side analysis identifies the structural mechanisms that would produce such compounding, the temporal rate of compounding observed in the pilot sample (n = 4) cannot be generalised across the broader enterprise population from this evidence alone. This contribution is treated as a candidate empirical claim warranting longitudinal validation.

Enterprises: Enterprises should evaluate AI platform adoption explicitly in terms of long-term dependency exposure rather than near-term capability acquisition alone. Strategic decisions should account for switching costs embedded in pricing models, technical architectures, ecosystem complementarities, and organisational learning processes. The structural inseparability of value capture and lock-in means enterprises cannot simultaneously maximise platform value and preserve full exit optionality; the strategic decision is therefore not whether to incur lock-in but where to incur it deliberately. Practical implications include investing in abstraction layers and portability-by-design where workloads warrant it, accepting deeper lock-in selectively for workloads where the platform's value advantage justifies it, and negotiating contract terms (commitment periods, exit clauses, data portability provisions) with explicit awareness of the structural switching costs that contracts cannot address. This awareness is increasingly visible in enterprise practice: industry analysis reports that approximately 15 percent of enterprises are shifting toward private AI deployments on private clouds in 2026, with rising AI costs, data lock-in, and operational risk identified as primary drivers (Forrester Research, 2025).

Platform Providers: Platform providers can sustain competitive advantage through ecosystem design and value creation rather than contractual restrictions. The most durable forms of lock-in are those generated through deep value integration, such as managed AI services, integrated MLOps environments, and ecosystem complementor density. They are more durable than deliberately restrictive mechanisms such as egress fees or proprietary data formats, which are increasingly subject to regulatory and competitive pressure. Providers seeking sustainable platform dominance should prioritise the depth and quality of value-creating features over the visibility of restrictive ones.

Policymakers: Regulatory approaches to AI infrastructure competition should move beyond portability and interoperability mandates to address structural and ecosystem-level sources of lock-in. The OECD (2025) identifies this as a domain where structural market features reinforce concentration in ways that interoperability requirements alone are unlikely to dissolve. The findings reinforce that identification empirically: contractual and data-format interventions address the visible surface of switching costs but leave the structural layer - value-embedded mechanisms, within-user learning dynamics, compliance infrastructure dependencies - largely untouched. Effective policy intervention will require new regulatory tools targeting these structural dimensions rather than refining the existing contractual ones. Industry projections that platform lock-in will rise from 5 percent to 35 percent of organisations by 2027 (Gartner, 2026) reinforce the time-sensitivity of this regulatory shift: the structural mechanisms identified here are intensifying faster than current regulatory instruments are evolving to address them. The UK Competition and Markets Authority's (2025) cloud services market investigation provides direct regulatory corroboration. The CMA concluded that competition in UK cloud services markets is not working well, that Microsoft and AWS hold „significant unilateral market power “with returns „substantially above their cost of capital... over a sustained period “(CMA, 2025, paras. 8, 36a), and that levels of concentration are likely to endure. The CMA illustrates that if cloud prices were 5% above competitive levels, UK customers would pay approximately £500 million more per year than in well-functioning markets (CMA, 2025, para. 37, n. 3). To remedy these adverse effects on competition, the CMA recommended designation of both Microsoft and AWS with strategic market status “under the UK's Digital Markets, Competition and Consumers Act, enabling targeted interventions that „directly benefit most UK customers and produce wider indirect effects by altering the competitive conditions for other providers “(CMA, 2025, paras. 38-39). The CMA's interventions, focused on structural rather than purely contractual remedies, align with the present study's argument that effective regulatory tools must address value-embedded and ecosystem-level switching costs rather than contractual surface mechanisms alone.

7.4. Limitations

Five limitations condition the findings of this study.

First, the demand-side empirical contribution is based on a small pilot sample of four informants across three industry sectors - telecommunications (two informants), federal services / management consulting (one informant), and financial services (one informant). While the sample meets the original sampling plan, it remains small relative to the diversity of enterprise AI adoption contexts and may carry self-selection effects favouring informants with pronounced views on platform dependency. The pilot sample provides initial evidence in support of analytical generalisation but does not establish saturated findings on enterprise strategic responses across the full range of sectors, organisational sizes, and decision-making contexts. Findings on enterprise behaviour should be treated as candidate patterns warranting further empirical validation. Sector-specific dynamics - and how they interact with the platform-generic mechanisms - are a particular dimension warranting further investigation.

Second, the single-case design supports analytical generalisation to theoretical propositions but does not provide statistical generalisation to a population of platforms or enterprises (Yin, 2014).

Third, the supply-side analysis is restricted to AWS and Azure as the dominant configuration of enterprise AI platform provision. Mechanisms specific to other platforms - Google Cloud, Oracle Cloud, IBM Cloud, regional sovereign clouds, or specialist AI infrastructure providers - lie outside scope and may exhibit different characteristics.

Fourth, the supply-side analysis relies on publicly available primary sources. Internal architectural details, undisclosed commercial terms, and provider-specific governance practices not publicly disclosed lie outside scope, which may understate the depth of additional mechanisms operating beneath public visibility.

Fifth, the findings are conditional on the current regulatory environment. Industry analysis suggests that digital sovereignty considerations - "geo-patriation," the relocation of workloads from global hyperscale's to regional or national alternatives - will become more prominent in enterprise platform decisions over coming years (Gartner, 2026), which may moderate the durability of some switching cost mechanisms identified here.

7.5. Future Research

Four directions for future research follow from the limitations of the current design.

First, comparative analyses extending beyond AWS and Azure to include Google Cloud, Oracle, IBM Cloud, and emerging providers (Alibaba Cloud, regional sovereign clouds) would establish whether the platform-generic mechanisms identified here are universal characteristics of enterprise AI infrastructure or specific to the AWS-Azure duopoly. The current two-unit design cannot distinguish between these possibilities.

Second, quantitative measurement of switching costs across enterprise AI platform transitions would establish empirical magnitudes for the mechanisms this study identifies qualitatively.

Third, sector-specific studies in additional regulated industries such as healthcare, Pharmaceuticals, Defence, Aviation would test the regulatory amplification dynamic identified in P6 with domain-specific empirical evidence.

Fourth, longitudinal research tracking enterprise AI adoption over time would allow direct observation of how switching costs compound through continued platform usage. Since P5 describes a temporal process, point-in-time evidence alone cannot fully confirm the dynamic.

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Appendix A: Demand Side Interview Guide

Interviewees were sent the following interview guides prior to the interviews:

About this study

This research examines how enterprise organisations experience and strategically manage the dependencies that arise from deep integration with AI infrastructure platforms such as Microsoft Azure and Amazon Web Services. As AI services become central to enterprise operations, the costs and complexities of changing providers - or maintaining flexibility across providers - are becoming strategically significant decisions for senior technology leaders.

The research combines analysis of publicly available platform documentation and pricing structures with conversations with senior enterprise technology leaders. Your experience of how these dynamics play out in practice is the primary purpose of this interview.

The research is conducted as part of an MBA dissertation and is not sponsored by or affiliated with any platform provider.

Confidentiality and your rights

Voluntary participation: Your participation is entirely voluntary. You may decline to answer any question and may withdraw at any time without consequence.

Confidentiality: Your organisation will be identified only by sector and size in any published output. You will be referred to by role category - for example, „Enterprise Architect, Financial Services" - and never by name.

Recording: With your permission, the interview will be recorded for transcription purposes only. The recording will not be shared with any third party and will be securely deleted after transcription.

Data storage: Your data will be stored securely and deleted when no longer required for this research.

Results: You are welcome to receive a summary of the findings once the research is complete. Please let the researcher know if you would like this.

This research is conducted in accordance with the research and ethics guidelines of the Blekinge Institute of Technology.

By proceeding with the interview, you confirm that your participation is voluntary and that you consent to the interview being recorded for transcription purposes.

How This Guide Works

This guide is provided in advance so you can see exactly what the conversation covers. It contains nine questions organised around the themes below. The interview is a conversation - not an examination. You are not expected to answer every question, and there are no right or wrong answers.

The conversation will follow your experience. If a particular question is not relevant to your context, we will move on. If one question opens into a rich discussion, we will stay there. Depth on the topics most relevant to your experience is more valuable than breadth across all nine questions.

The nine questions cover the following themes:

Q1 - How and when your AI integration became difficult to reverse

Q2 - Which systems depend on the provider and what switching would involve

Q3 - How much of your data and AI assets are portable versus platform-specific

Q4 - Which capabilities are most valuable - and whether those are the same ones creating the deepest dependency

Q5 - Whether you have ever chosen a less capable option to avoid lock-in

Q6 - How your negotiating power has changed and whether you have an exit strategy

Q7 - Whether regulatory requirements are helping or worsening your lock-in situation

Q8 - The reality of multi-provider AI strategy in practice

Q9 - Your forward exposure if your provider made a major change

Opening

Q1
Integration history and the lock-in threshold
How has your AI stack been integrated into your core workflows over the past 12-24 months - and at what point did you feel the integration became difficult to reverse? Was there a specific decision or deployment that marked that threshold, or did it happen gradually?

Dependency and switching experience

Q2
System dependency and switching reality
Can you walk me through which internal systems now depend on the AI provider's APIs, SDKs, or proprietary data formats - and have you ever seriously explored migrating away from them? What barriers did you encounter - technical, organisational, or contractual - and which proved most significant in practice?
Q3
Data asset portability
How much of your proprietary data - fine-tuning datasets, embeddings, model weights, agent configurations - lives inside the provider's infrastructure? Have you ever attempted to extract or replicate it elsewhere, and do you have a clear picture of what that would realistically involve?

Value capture and the lock-in-trade-off

Q4
Capability value versus dependency depth
Which AI capabilities from your current provider would you most struggle to replicate elsewhere - and are those the same capabilities that are most central to your day-to-day operations? Is there a pattern where the features you rely on most are also the ones binding you most tightly?
Q5
Choosing less capability to preserve flexibility
Have you ever chosen a slightly inferior AI capability because switching away from a superior one would have required replacing other parts of your enterprise stack? If so, how did that trade-off get made - was it a formal decision or an emergent one?
Q6
Negotiating leverage and exit planning
How has your ability to negotiate pricing, SLAs, or data terms changed as your dependency on the provider has deepened - and at what point did you feel the balance of leverage shift? Does your organisation have a formal AI vendor governance or exit strategy, and if so, has it ever been invoked?

Regulatory dimensions

Q7
Regulation as portability driver or lock-in amplifier
Are compliance requirements - GDPR, the EU AI Act, data residency rules, sector-specific regulations - pushing your organisation toward portability, or are they making lock-in worse? Do your audit trail and compliance infrastructure depend on provider-specific tooling?

Multi-provider reality and forward exposure

Q8
Multi-provider strategy in practice
In practice, do you actually run multiple AI providers, or does it mainly sound good strategically? Which workloads sit where and why - and what is the real operational cost of maintaining that flexibility versus going deep with one provider?
Q9

Forward vulnerability and board visibility
If your current provider announced a 50% pricing revision, a major model deprecation, or a sustained outage next quarter - what specifically breaks in your operations, how long would it take to recover, and what does that tell you about where your real exposure lies? Is that level of exposure visible to your board or executive team?

Closing

Those are all nine questions. Before we finish, two brief closing questions:

- Is there anything you expected me to ask that I did not - something about your experience of AI platform dependency that feels important and has not come up?
- Are there colleagues in your network - in similar roles at other organisations - who you think would have valuable perspectives on these questions and might be willing to speak with me?

Thank you very much for your time. Your contribution to this research is genuinely appreciated. If you have any questions about the study or how your data will be used, please contact me at [Email]

I am happy to share the findings with you once the research is complete - just let me know if you would like that.

Clement Nkamanyi / Devakumar Kannan

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Appendix B: Demand Side Request for interview Email template

Dear [Name],

Thank you for agreeing to participate in our research study on enterprise AI platform strategy. I genuinely appreciate your willingness to contribute your experience and perspective.

Please find attached the Interview Discussion Guide for our upcoming conversation. The guide contains nine questions organised around the following themes:

1. How and when your AI integration became difficult to reverse
2. Which systems depend on the provider and what switching would involve
3. Data and AI asset portability
4. The relationship between capability value and platform dependency
5. Governance, negotiating power, and exit strategy
6. Regulatory dimensions of lock-in
7. The reality of multi-provider AI strategy in practice
8. Your organisation's forward exposure

A few things worth noting before our conversation:

The guide is provided for your preparation - you are not expected to answer every question, and there are no right or wrong answers. The conversation will follow your experience and I will focus on the areas most relevant to your context.

All responses will be treated in strict confidence. Your organisation will appear only as a sector and size category in any published output, and you will be referred to by role category rather than by name.

The interview will last approximately 60 minutes and will be conducted remotely via video call.

To schedule our conversation, could you please share one or two dates and times that work for you over the coming days? I will confirm one and send a calendar invitation promptly.

If you have any questions about the guide or the research before we speak, please do not hesitate to reach out.

I look forward to our conversation.

Best regards,

Appendix C: Table references

Table 1 - Lock-in mechanism to proposition map

Mechanism	**AI Stack Layer**	**Switching Cost Type**	**Enterprise implication**	**Proposition (P#)**	**Source anchors**
Managed AI services and proprietary APIs	Platform / Model	Compatibility, learning	Reduced portability; workflow re-engineering on migration	P1, P2	Hagiu and Wright (2025); Besanko et al. (2017)
Marketplace ecosystems and ISV dependencies	Application	Ecosystem loss	Complementor dependency; non-transferable network value	P1, P3	Rong et al. (2026); Costabile (2024); Chen et al. (2022)
Data abstraction and proprietary storage	Data	Migration, retraining	Sunk investment; data gravity trap	P1, P2	Hagiu and Wright (2025); Besanko et al. (2017)
Usage-based and commitment pricing structures	Governance	Financial	Exit penalties; multi-homing disincentives	P3, P4	Farrell and Shapiro (1988); He et al. (2023)
Value-creating features as lock-in mechanisms	Platform / Model/ Application	Compatibility, financial, learning	Value capture and dependency exposure inseparable	P2, P4	Besanko et al. (2017); Evans and Schmalensee (2016); Rong et al. (2026); Jacobides et al. (2018)
Within-user learning and model customization	Platform / Model	Learning, sunk cost	Autonomous lock-in deepening through normal usage	P5	Hagiu and Wright (2025); Swann (2009)

Compliance infrastructure and audit tooling	Governance / Regulatory	Organizational, financial	Regulatory amplification of structural dependency	P6	OECD (2025); Opara-Martins et al. (2016)

Table 2 - Embedded Unit Structure (adapted from Yin, 2014, p. 50)

Unit Type	Instances	Research Question	Analytical Role
Type A (Supply Side)	Amazon Web Services; Microsoft Azure	RQ1: What architectural and governance mechanisms create switching costs?	Characterise the mechanisms through which AI platforms generate dependency
Type B (Demand Side)	Enterprise organisations operating within these ecosystems	RQ2: How do enterprise firms navigate switching cost exposure in enterprise AI infrastructure platforms?	Document strategic responses, mitigation choices, and governance decisions

Table 3 - Multi-Source Evidence

Unit	Data Type	Sources	Purpose
Type A	Architecture documentation & whitepapers	AWS Well -Architected Framework; Azure Architecture Centre; AI/ML service docs	Identify architectural mechanisms generating switching costs (RQ1)
	Analyst reports & Institutional reports	Gartner (2025,2026), Forrester (2025), Synergy Research Group (2026); OECD (2025); Competition and Markets Authority (2025)	Corroborate governance and ecosystem mechanisms (RQ1)
Type B	Semi-structured interviews	3 semi-structured pilot interviews (target was 4-5) with senior technology leaders, including VP of AI	Primary data on strategic management of

		Transformation, Head of AI Transformation, and Senior Enterprise Solutions Architect.	switching costs (RQ2)

Table 4 - Operationalization table

Construct	**AI-Platform Manifestation**	**Empirical indicator / Analytical code**	**Data source**	**Proposition**
Compatibility switching cost	Proprietary MLOps toolchains (SageMaker, Azure ML); platform-native AI APIs	Proprietary API dependency, workflow re-engineering effort on migration	Architecture documentation; interviews	P1, P2
Learning switching cost	Developer certification specificity; accumulated organisational know-how; within-user learning effects	Certification specificity, accumulated organizational know-how, developer toolchain familiarity	Interviews, analyst repots	P1, P5
Financial switching cost	Enterprise Discount Programmes; committed-use contracts; data egress pricing; model fine-tuning sunk investment	Contract duration; minimum spend; egress fee schedules; forfeiture terms; commitment forfeiture	Pricing schedules; contractual terms; interviews	P3, P4
Ecosystem switching cost	ISV marketplace dependencies; complementor network non-transferability; bundled service integration	Marketplace integration value; complementor	Architecture documentation; interviews	P1, P3

		lock-in; non-transferable network value		
Within-user learning effect	Platform customisation depth over time; model fine-tuning investment; prompt library accumulation	Autonomous lock-in compounding through normal usage	Interviews; analyst reports	P5
Asset specificity (TCE)	Platform-specific workflow encoding; proprietary runtime dependency; compliance infrastructure entrenchment	Relationship-specific investment that loses value upon switching entrenchment	Architecture documentation; interviews	P4, P5
Ecosystem governance mechanism	Pricing tier structure; access conditions; marketplace curation policy; multi-homing pricing disincentives; partner programme terms	Governance design as dependency instrument; pricing structures; partner programme terms	Pricing schedules; policy documents; analyst reports	P3, P6
Regulatory amplification	Compliance infrastructure dependency; data residency entrenchment; audit trail lock-in	Regulation as lock-in amplifier rather than constraint	Contractual terms; analyst reports; interviews	P6

Table 5 - Informant coding reference

Code	**Role**	**Sector**	**Experience**	**Interview length**
VP-Telco	Vice president, AI Transformation (R&D)	Telecommunications	25+ years	~60 minutes

Head-Telco	Head, AI Transformation	Telecommunications	15+ years	~55 minutes
Architect-FedSvc	Senior Enterprise Solutions architect	Federal Services / Management consulting	24+ years	~65 minutes
Head-FinSvc	Head of AI	Financial Services	23+ years	~45 minutes